

A Statistical Model of Auroral Ion Precipitation

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The average global characteristics of precipitating auroral ions were determined using the data from the SSJ/4 detectors on the F6 and F7 satellites of the Defense Meteorological Satellite Program (DMSP). For this study the high-latitude region was divided into spatial elements in magnetic local time (MLT) and corrected geomagnetic latitude (CGL). One such spatial matrix was created for each of seven levels of magnetic activity as defined by K_p . Approximately 26.5 million, individual, 1-s spectra were used to determine the average ion spectrum over the energy range from 30 eV to 30 keV in each spatial zone and at each level of activity. Where appropriate, the spectra were extrapolated to 100 keV to provide a more complete estimate of the total integral energy flux, number flux, and average energy. The global patterns of the integral energy flux, integral number flux, and average energy derived from the average spectra vary smoothly with latitude, MLT, and activity. For a given K_p value the higher levels of integral energy flux occur in C-shaped regions symmetric about a meridian running prenoon to premidnight. The maximum integral energy flux is found premidnight; the minimum peak integral energy flux in latitude is found prenoon. Except in the prenoon region, the overall level of integral energy flux increases with K_p with the premidnight maximum increasing by a factor of 6.1 from $K_p = 0$ to $K_p \geq 6$. The maximum integral number flux is centered at noon and is taken to be the center of the cusp. The maximum value does not vary significantly with activity, although its location moves to lower latitude with increasing K_p . A minimum in the average energy also occurs in the noon meridian, several degrees poleward of the integral number flux maximum. The separation in latitude between the integral flux maximum and the average energy minimum of the cusp increases with K_p . The maximum average energy occurs on the eveningside of the oval near the equatorward boundary of the ion precipitation and shifts toward noon with increasing activity. The average integral number flux for the precipitating auroral ions is 1-2 orders of magnitude less than that for the precipitating auroral electrons at all latitudes, MLTs, and activities. The integral energy flux of the ions, on the eveningside of the oval, can equal or exceed that for the electrons near the equatorward edge of the auroral region. Integrated globally, the ratio of the ion to electron number flux varies from 0.024 to 0.015 over the range of activity from $K_p = 0$ to $K_p \geq 6$. The ratio for the integral energy flux ranges from 0.11 to 0.17 with no trend in activity.

1. INTRODUCTION

The existence of proton precipitation in the auroral regions was first inferred from ground-based observations of Doppler-shifted hydrogen Balmer series emissions [Meinel, 1950]. These and other such ground-based measurements are responsible for much of what is known about the general morphology of low-altitude ion precipitation. They have shown that the proton precipitation occurs within an annular region roughly collocated with the region of electron-produced diffuse and discrete auroral forms. Within this region the inferred intensity of the precipitation decreases toward noon [Eather, 1967, and references therein] and on the eveningside (morningside) of the oval the proton precipitation typically extends to latitudes below (above) the region of electron precipitation. The entire pattern moves equatorward and intensifies with increases in the level of geomagnetic activity and substorm activity [Montbriand, 1971; Fukunishi, 1975; Vallance Jones *et al.*, 1982]. Much of the hydrogen emission seen in the dayside auroral region is taken to result from plasma injected at midnight that is subsequently transported to the dayside by the convection and gradient and curvature drift of the particles [Eather *et al.*, 1976].

Direct measurements of the precipitating ions at low altitudes have confirmed and expanded the picture of the general morphology derived from the optical observations [Hultqvist, 1979, and references therein]. A series of these

observations have shown that in the nightside auroral region, during quiet times the latitudinal variation in ion characteristics is consistent, over a range of latitudes, with adiabatic compression of the plasma as it is convected toward the Earth from the tail and the possible formation of Alfvén layers nearer to the Earth [Hultqvist, 1974; Galperin *et al.*, 1976, 1978; Hultqvist, 1979, and references therein; Sauvaud *et al.*, 1981]. These observations are at some variance with measurements made at higher altitudes that show the inner magnetosphere often to be populated by the sudden injection of hot plasma [McIlwain, 1974]. Researchers have hypothesized that during such injections the adiabatic invariants for the particles may be violated. A great deal of analysis has been done on the earthward boundary of such hot plasma in the inner magnetosphere to determine whether the boundary arises due to the establishment of steady state Alfvén layers [Kivelson *et al.*, 1985], the combination of Alfvén layers with a time varying electric field [Kaye and Kivelson, 1979; Chiu and Kishi, 1984], particle injection to a double spiral injection boundary [Konradi *et al.*, 1975; Mauk and Meng, 1983a, b, 1986] or traveling injection fronts [Moore *et al.*, 1981].

In the dayside auroral region, precipitating ions with magnetosheath characteristics are observed at low altitudes within the cusp and cleft [Heikkilä and Winningham, 1971]. For a southward B_z component of the interplanetary magnetic field (IMF), these precipitating ions display a dispersion in energy with latitude across the cusp consistent with entry of the plasma over a limited spatial region and the subsequent convection of the particles by the magnetospheric electric field. For northward B_z , energy dispersions

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are observed that are interpreted as the signature of diffusive processes populating the cusp [Reiff *et al.*, 1977, 1980; Reiff, 1979; Shelly *et al.*, 1979; Burch *et al.*, 1980].

The above cited research was in the form of case studies or relatively small-sample, statistical analyses. To date, there has been only one study where the average global pattern of auroral ion precipitation was determined [Reidler, 1972; Reidler and Borg, 1972]. This work used the ion data from the ESRO satellite. The study was limited by the relatively small data sample available and the restricted spectral information provided by the ESRO ion instrument. Such global studies are important in that they provide a global picture of the contribution of the ions to the dynamics of the auroral region.

In order to better assess this contribution we have undertaken a statistical analysis similar to that of Reidler [1972] and Reidler and Borg [1972] using the ion data from the SSJ/4 detectors on the F6 and F7 satellites of the Defense Meteorological Satellite Program (DMSP). In the study we have determined the average ion spectrum as a function of magnetic latitude, magnetic local time and geomagnetic activity as measured by Kp . This data set is particularly well suited for a statistical study because of its size (many tens of millions of data points) and the relatively high sensitivity of the SSJ/4 detector compared to other ion detectors that have been flown at low altitude.

This paper is divided into six sections. Following the introduction, section 2 gives the instrumentation and data coverage. In section 3 the binning procedure for the statistical study is described. Section 4 lists the results. Section 5 is the discussion and section 6 is the conclusions. The appendix covers details of the extrapolation of the ion spectra.

2. INSTRUMENTATION AND DATA COVERAGE

The data for this study are from the SSJ/4 detectors flown on the F6 and F7 satellites of DMSP. Each SSJ/4 detector consists of an array of four cylindrical, curved plate, electrostatic analyzers configured into two pairs; one pair each to measure electrons and ions. Each pair covers the energy range from 30 eV to 30 keV in 20 channels spaced at equal intervals in energy on a logarithmic scale. In each pair, one sensor covers the energy range from 30 eV to approximately 1 keV in 10 channels and the other, the range from 1 keV to 30 keV, also in 10 channels. The two sensors in each pair are stepped together such that a complete 20-point spectrum is returned once per second. The sensors have peak geometric factors of 2×10^{-2} (2×10^{-4}) and 8.8×10^{-4} (8.8×10^{-4}) $\text{cm}^2 \text{sr}$ for the low- and high-energy sensors for ions (electrons) respectively and $\Delta E/E$ of approximately 0.1 for both sensors and for both electrons and ions. The geometric factors for the ion sensors are approximately a factor of 10 to a factor of 100 larger than those for typical detectors flown previously in the low-altitude polar environment (for more information see Hardy *et al.* [1984]).

The DMSP F6 and F7 satellites are in circular, 840-km, sun-synchronous orbits at an inclination of 98.3° . The DMSP F6 satellite was launched in November 1982 into the dawn-dusk meridian. The DMSP F7 satellite was launched in December 1983 into the 1030–2230 meridian. Both satellites are three axis stabilized, and the SSJ/4 sensors are configured on the spacecraft such that their look directions are toward the local vertical at all times. The SSJ/4 data from the

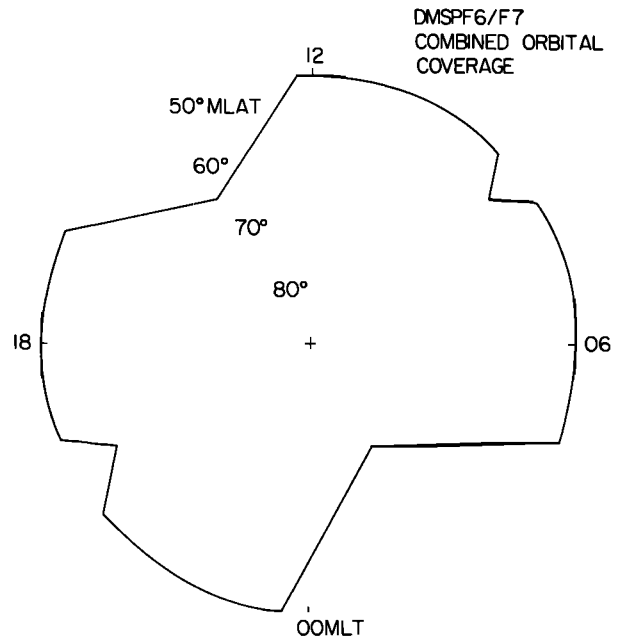


Fig. 1. The orbital coverage of the combined DMSP F6 and F7 satellites for both north and south poles in a corrected geomagnetic latitude-magnetic local time coordinate system.

two satellites are used operationally by the Air Force Global Weather Central to specify the level and extent of auroral precipitation. As such, the SSJ/4 detectors are operated continuously, and the data are returned and recorded for approximately 90% of the time.

Although both satellites are in sun-synchronous orbit, they cover a wide range in corrected geomagnetic latitude (CGL) and magnetic local time (MLT) due to the offset between the geographic and geomagnetic poles. In Figure 1 the combined orbital coverage of the two satellites across both hemispheres in corrected geomagnetic latitude and magnetic local time is shown. For latitudes above 50° the coverage is nearly complete except for a small region post-midnight and postnoon.

3. DATA BINNING PROCEDURE

In this study we followed much the same procedure as in our previous work on the average pattern of electron precipitation [Hardy *et al.*, 1985]. The high-latitude region above 50° was divided into a series of spatial bins in CGL and MLT. There were 30 bins in CGL and 48 one-half-hour bins in MLT. In latitude the bins were 2° wide between 50° and 60° and 80° and 90° and 1° wide between 60° and 80° . One such spatial matrix was created for each whole number of Kp , i.e., one for $Kp = 0, 0+$, one for $Kp = 1-, 1, 1+$, etc., up to $Kp = 5-, 5, 5+$. One final matrix was created for all cases where $Kp \geq 6-$.

Using a large sample of data from the SSJ/4 sensors on the DMSP F6 and F7 satellites, the average spectrum in each spatial element and at each level of activity was determined. Altogether 1 year of data was used from each satellite (all of 1983 for F6 and all of 1984 for F7). This data set comprises approximately 18,000 passes over the northern or southern hemisphere and 26.5 million individual spectra. The data were divided by Kp as follows: $Kp = 0$, 2.67%, $Kp = 1$, 14.1%, $Kp = 2$, 26.2%, $Kp = 3$, 26.4%, $Kp = 4$, 18.4%, $Kp = 5$, 8.66%, $Kp \geq 6-$, 3.07%.

Initially, the data for the two satellites were binned separately. Once the binning was complete, a set of simple algorithms was applied to subtract from the data the low level of background produced principally by penetrating high-energy particles that by directly stimulating the channeltrons in the sensors produce counts. This background was confined generally to latitudes below the auroral zone where the counts are produced primarily by high-energy particles from the outer zone of the Earth's radiation belts. Although the background varies with latitude and MLT, the background level can be determined since, in this region, such particles produce, on average, for a fixed latitude and MLT, the same count rate in all or most of the channels of the detector. In a small number of MLTs, this background overlapped 1° – 2° into the region of auroral fluxes and for such cases this background was subtracted. For the vast majority of MLTs, latitudes and Kp no background of this type was subtracted in the auroral zone and polar cap. Noise levels in these regions are estimated to be less than 0.2 counts/s in most channels. This is orders of magnitude less than the typical average count rates in these regions and therefore has a negligible effect on the quantities calculated. There were no solar flares in the period used in this study that produced a significant background in the instruments.

As the next step, the average spectra determined from each satellite data set were visually inspected to remove any spike produced by telemetry noise that was not caught in the original binning of the data. In all, only approximately 10 out of the roughly 10,000 average spectra were corrected in this way. For spatial zones where such spikes were observed and average spectra were available from both satellites, the average spectrum from the satellite data containing the spike was eliminated. For other cases the spectrum was eliminated and then reconstructed by linearly interpolating in latitude and MLT.

As a final step, the data from the two satellites were cross normalized. This was done by calculating the average count rate spectra, summed over Kp , for each satellite over the latitude and local time bins where the coverage of the two satellites overlapped. Taking the ratio of the count rates for the two average spectra then gives the relative response of the two instruments.

The results of this procedure are shown in Figure 2. For the high-energy channels the ratio is approximately constant with a value of 0.68. For the low-energy channels the ratio is near one at 1 keV, increases linearly to a value of 1.35 at 44 eV, and then decreases to 0.64 for the lowest-energy channel. F7 was chosen as the standard since it had the better ground calibration. The average count rates for F6 were multiplied by the reciprocal of the ratios shown in Figure 2 and the weighted mean of the count rates of the two detectors determined. Analysis of concurrent data from the two satellites verifies that this was not a solar cycle induced effect.

Inspection of the final average spectra showed that a significant portion of the ion spectrum in many latitude and local time bins was at energies above 30 keV. In order to include the contribution to the auroral flux above 30 keV, some of the average spectra were extrapolated to 100 keV. There is obviously some uncertainty in such an extrapolation. Nonetheless, we believe the extrapolation to be reasonable since, as will be shown in the paper, these extrapolations begin at energies well beyond the peak in the

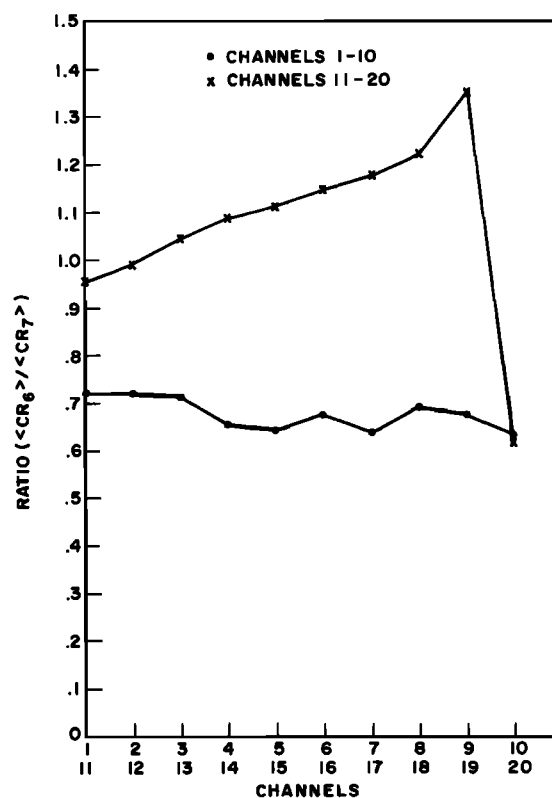


Fig. 2. The ratio of the average count rates in the 20 channels of the SSJ/4 detector on the DMSP F6 satellite to the average count rates in the 20 channels of the SSJ/4 detector on the DMSP F7 satellite summed over all activities and for all regions where the orbital coverage of the two satellites overlap. The channels are numbered 1–20 with channel 1 being the highest-energy channel and channel 20 the lowest-energy channel.

differential number flux spectrum; that is, we are extrapolating the high-energy tail of the ion distribution. We maintain that the extrapolation is necessary because for many spectra, failure to extrapolate results in values of the integral flux, energy flux, and average energy strongly aliased by the energy range of the SSJ/4 detector. Further details on the extrapolation procedure, its effects and justification are contained in the appendix.

In the remainder of this paper we will discuss the results of this statistical study. Three principal forms will be used to display the results. First, we will use plots of the average spectra. Second, the average spectra are integrated to determine the average integral energy flux JETOT, the average integral number flux JTOT, and the average energy EAVE of the ions in each spatial element. The three quantities are calculated using

$$\begin{aligned}
 \text{JETOT} = & E_1 j(E_1)(E_2 - E_1) \\
 & + \sum_{i=2,22} E_i j(E_i) [(E_{i+1} - E_{i-1})/2] \\
 & + E_{23} j(E_{23})(E_{23} - E_{22})
 \end{aligned} \quad (1)$$

$$\begin{aligned}
 \text{JTOT} = & j(E_1)(E_2 - E_1) \\
 & + \sum_{i=2,22} j(E_i) [(E_{i+1} - E_{i-1})/2] \\
 & + j(E_{23})(E_{23} - E_{22})
 \end{aligned} \quad (2)$$

TABLE 1. Properties of the Ion Cusp Number Flux Maximum

Kp	MLAT, deg	JTOT, ions/cm ² s sr	JETOT, keV/cm ² s sr	EAVE, keV
0	79.5	1.34×10^7	1.59×10^7	1.19
1	78.5	1.03×10^7	1.47×10^7	1.43
2	77.5	1.13×10^7	1.78×10^7	1.57
3	76.5	1.25×10^7	2.69×10^7	2.15
4	75.5	9.71×10^6	2.19×10^7	2.26
5	72.5	1.40×10^7	3.19×10^7	2.28
≥ 6	69.5	1.05×10^7	2.59×10^7	2.47

$$\text{EAVE} = \text{JETOT}/\text{JTOT} \quad (3)$$

where $j(E_i)$ is the differential ion number flux in units of ions/cm² s sr eV for the i th energy channel and E_i is the central energy. For a Maxwellian distribution, EAVE as calculated above is equal to $2kT$ [Robinson *et al.*, 1987]. The three quantities are given in units of keV/cm² s sr, ions/cm² s sr, and keV, respectively. These values are displayed as histograms as a function of corrected geomagnetic latitude along the noon-midnight and dawn-dusk meridians or as color-coded polar spectrograms in a MLT-CGL coordinate system. In all cases the extrapolated spectra are used.

The values obtained for the integral energy flux, integral number flux, and average energy are smoothed slightly to eliminate variations due to the small level of statistical fluctuations and noise remaining in the data. The smoothing is performed by recalculating the value in each spatial element according to

$$a_{i,j} = (3a_{i,j} + a_{i+1,j} + a_{i-1,j} + a_{i,j+1} + a_{i,j-1})/7 \quad (4)$$

where $a_{i,j}$ refers to the value of the integral number flux, integral energy flux, or average energy in the i th latitude sector and the j th magnetic local time sector. One iteration of this procedure was performed on the data.

4. RESULTS

In this section we discuss the results of the statistical study. The section is divided into two parts. In the first part the maps of the average values of the integral number flux, integral energy flux, and average energy are discussed along with the shape of the average spectra and their variation in latitude and magnetic local time. In the second part, the average pattern of ion precipitation is compared to that for the electron precipitation.

Ion Characteristics

Global maps. In Plates 1a, 1b, 2a, 2b, 3a, and 3b the integral number flux, integral energy flux, and average energy, respectively, for the ions are plotted in a color polar spectrogram format for each of the seven levels of activity used. The plots illustrate the following:

1. The maximum integral number flux generally occurs on the dayside of the oval with the highest levels confined to a relatively small region in latitude as well as MLT. We identify the position of the number flux maximum as the center of the cusp. In Table 1 the characteristics and latitude of the cusp center are listed. The integral number flux at the maximum has no consistent trend to increase or decrease with increasing Kp . At all levels of activity the maximum on the dayside has a value of $1.2 \times 10^7 \pm 20\%$ ions/cm² s sr and

is centered on 1200 MLT. The integral energy flux and average energy both increase with increasing Kp , each by approximately a factor of 2. The integral number flux maximum moves to lower latitudes with increasing Kp from a position at 79.5° CGL for $Kp = 0$ to 69.5° CGL for the $Kp \geq 6$ case. The extent of this shift is in agreement with the motion of the electron integral number flux maximum reported in our previous study [Hardy *et al.*, 1985] and other earlier studies of the cusp/cleft motion as determined primarily from electron measurements and made over a relatively wide range of MLTs across the dayside oval [Burch, 1972; Yasuhara *et al.*, 1973; Carbary and Meng, 1986a, b]. The area of elevated values of the ion integral number flux defines a roughly circular or oval-shaped region centered about noon. The shape is consistent with the mapping to low altitudes of the funnel-shaped region of the magnetic field in the cusp as shown by Hedgecock and Thomas [1975].

2. Away from the maximum the peak level of integral number flux on the dayside is asymmetric in MLT with a higher level observed prenoon at all levels of activity.

3. A weak enhancement in the integral number flux is seen at subauroral latitudes in the 0600–1200 MLT quadrant. This is a spacecraft-induced effect. The solar cells of the DMSP satellite are configured such that positive leads are exposed to space. This causes the spacecraft to charge when the solar panels are illuminated. The maximum negative voltage produced on the spacecraft in this manner is the negative bus voltage of -28 V. Such negative voltage accelerates the ambient thermal ions around the satellite and can produce counts in the 30-eV channel of the detector. Because of minor differences in the average bus voltage for the two satellites and minor differences in the central energy of the lowest ion energy channel in the two SSJ/4s, the effect is slightly higher on average in the F7 than in the F6 data.

4. On the nightside of the oval the increase in the peak value of the integral number flux with activity is relatively small (on average approximately 20% for each increase in Kp by 1) but the oval width increases systematically with increasing activity. In Table 2 the characteristics and location of the maximum integral number flux along the midnight meridian are listed. The integral number flux maximum varies by a factor of approximately 2.5, from 3.1 to 7.9×10^6 ions/cm² s sr from lowest to highest activity. The location of the maximum moves 1° equatorward for each whole value increase in Kp . Except for the $Kp = 5$ case, the average energy of the ions at the maximum increases with increasing activity; from 7.7 keV for $Kp = 0$ to 19.2 keV for $Kp \geq 6$.

5. The maximum integral energy flux is always found on the nightside. Highest values of the integral energy flux occur within roughly C-shaped regions that are symmetric about a meridian running premidnight to prenoon. This is in

TABLE 2. Properties of the Midnight Number Flux Maximum

Kp	MLAT, deg	JTOT, ions/cm ² s sr	EAVE, keV
0	68.5	3.06×10^6	7.70
1	67.5	4.11×10^6	9.99
2	66.5	4.57×10^6	13.4
3	65.5	5.71×10^6	15.6
4	64.5	6.35×10^6	18.3
5	63.5	6.37×10^6	17.0
≥ 6	62.5	7.93×10^6	19.2

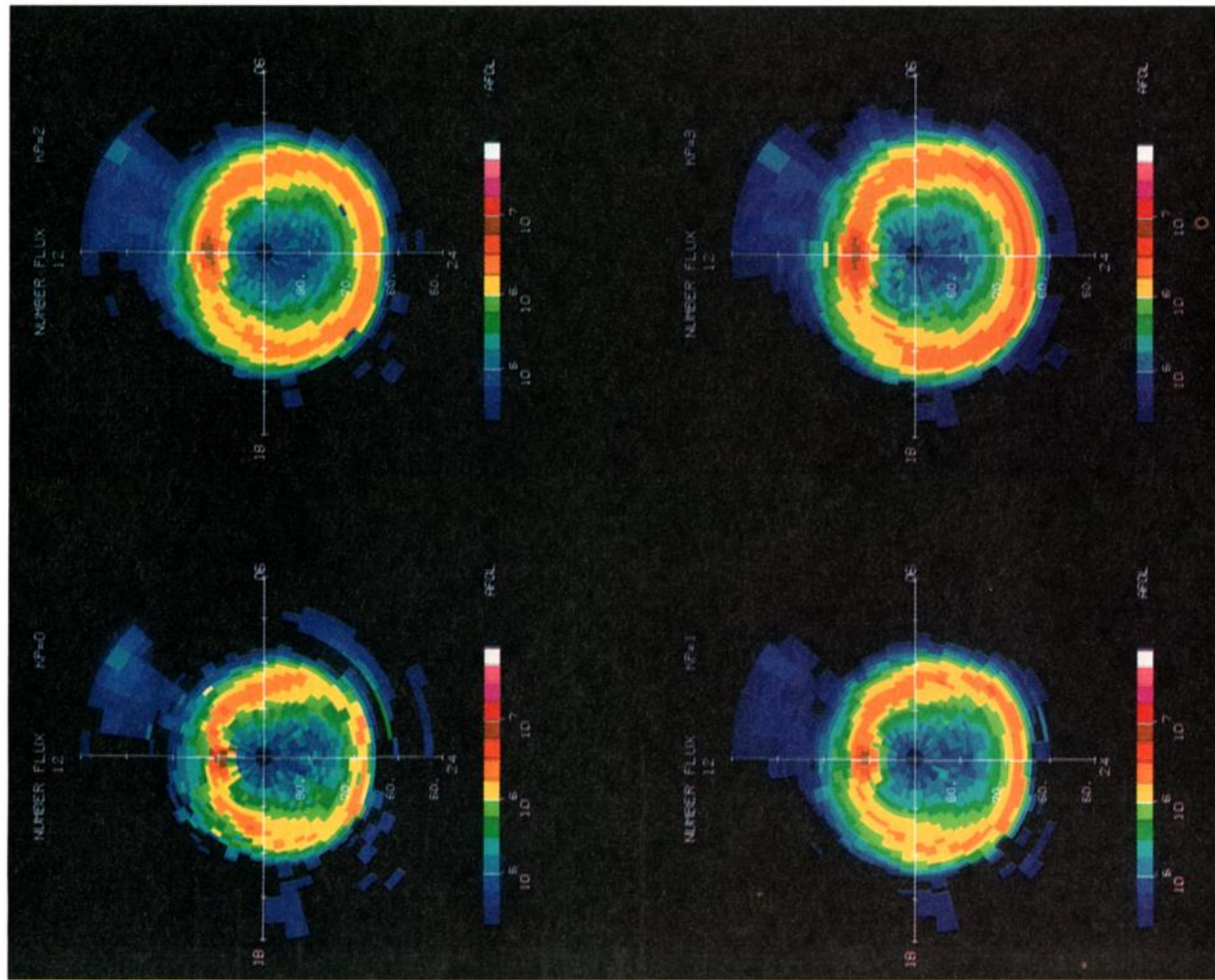


Plate 1a

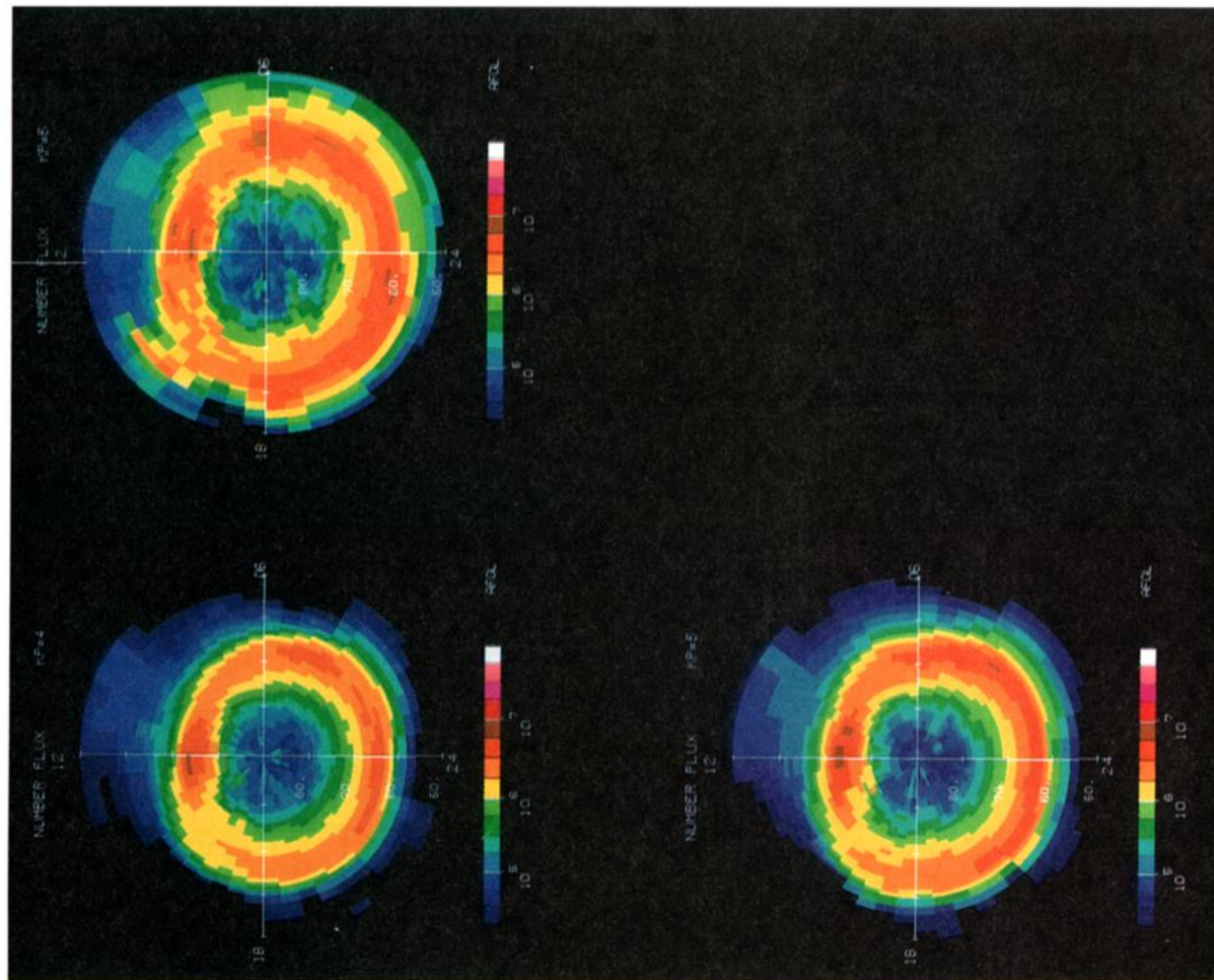


Plate 1b

Plate 1. Color polar spectrograms of the average integral number flux for each whole value of Kp from 1 to 5. The final spectrogram is for $Kp \geq 6$. The spectrograms are plotted in a magnetic local time-corrected geomagnetic latitude coordinate system. The values displayed are in units of $\text{ions}/\text{cm}^2 \text{ sr}$. The scale is logarithmic with four color division per decade change in the number flux.

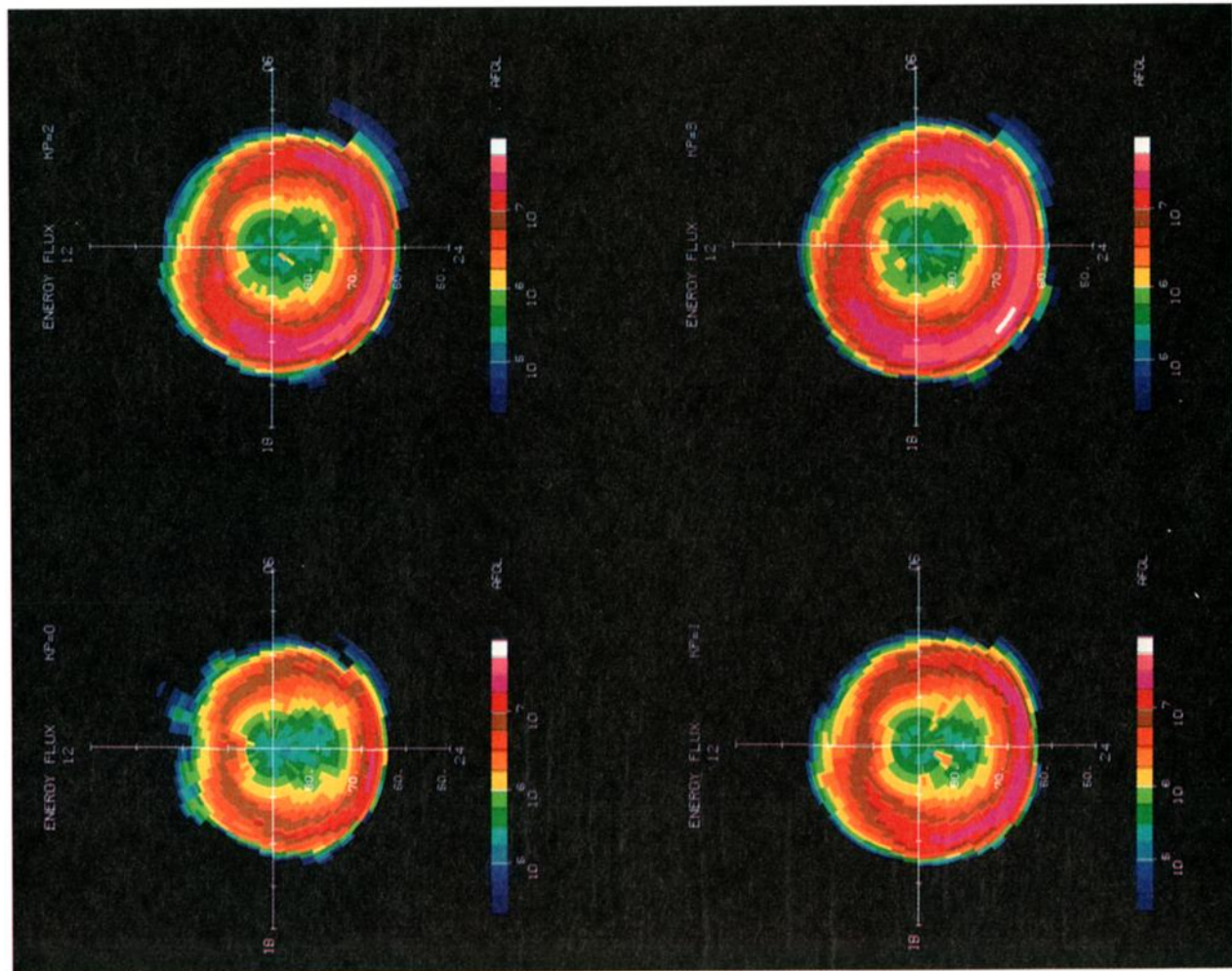


Plate 2a

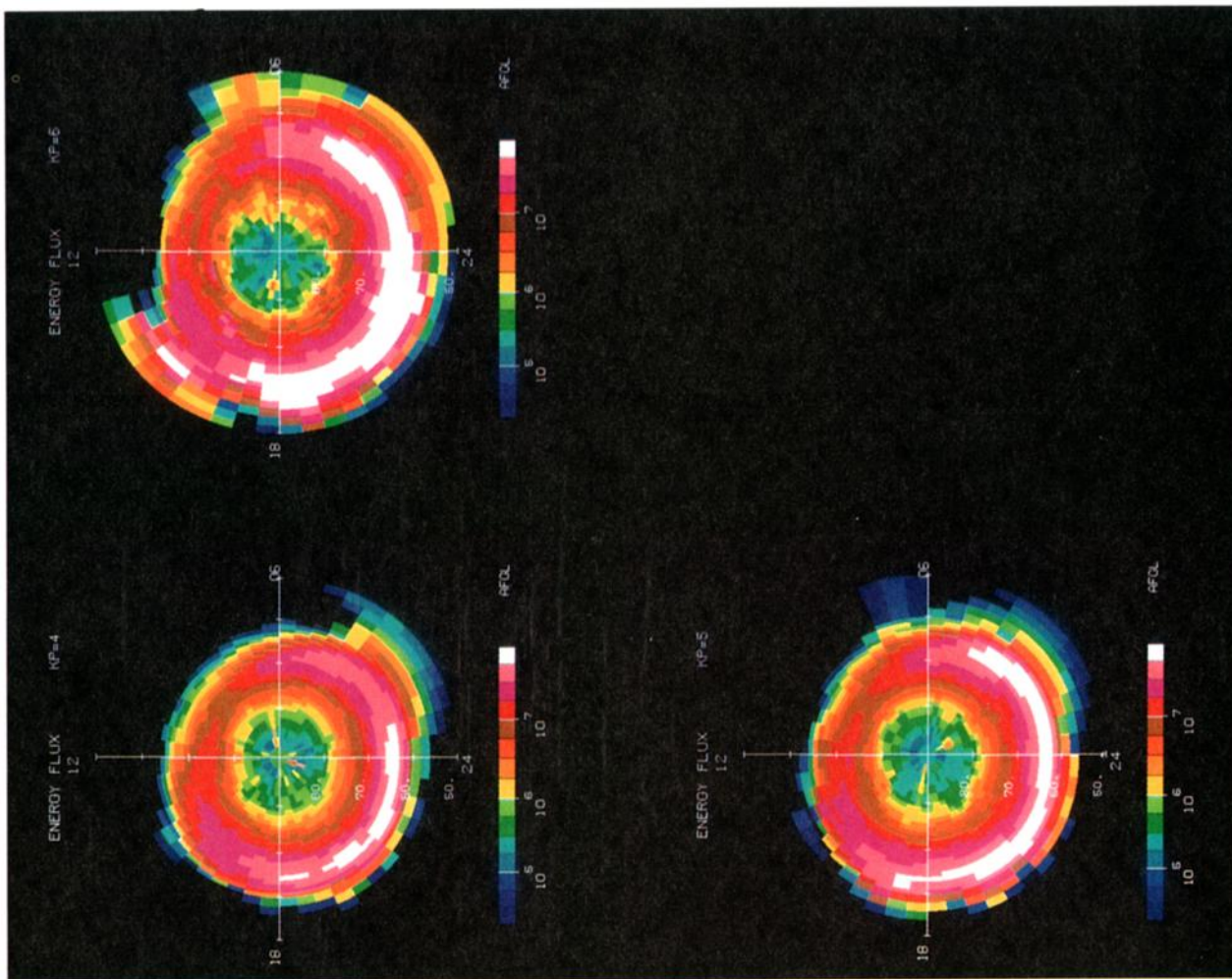


Plate 2b

Plate 2. Same as Plate 1 but for the integral energy flux in units of $\text{KeV}/\text{cm}^2 \text{ s sr}$.

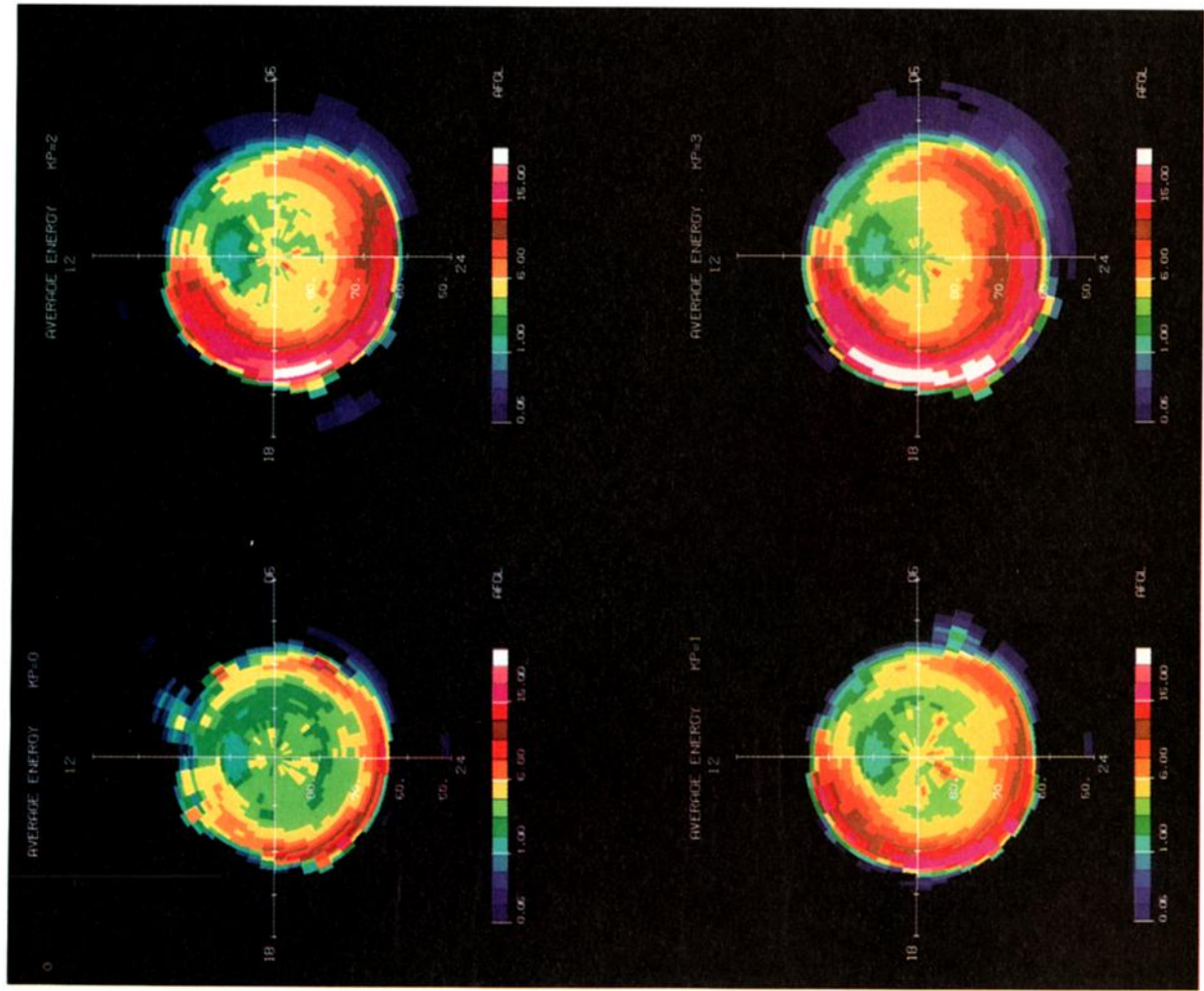


Plate 3a

Plate 3. Same as Plate 1 but for the average energy. The color scale corresponds to energies of 0.05, 0.1, 0.2, 0.5, 1.0, 2.0, 3.0, 4.0, 6.0, 8.0, 10.0, 12.0, 15.0, 20.0, 25.0, and 30.0 keV.

Plate 3b

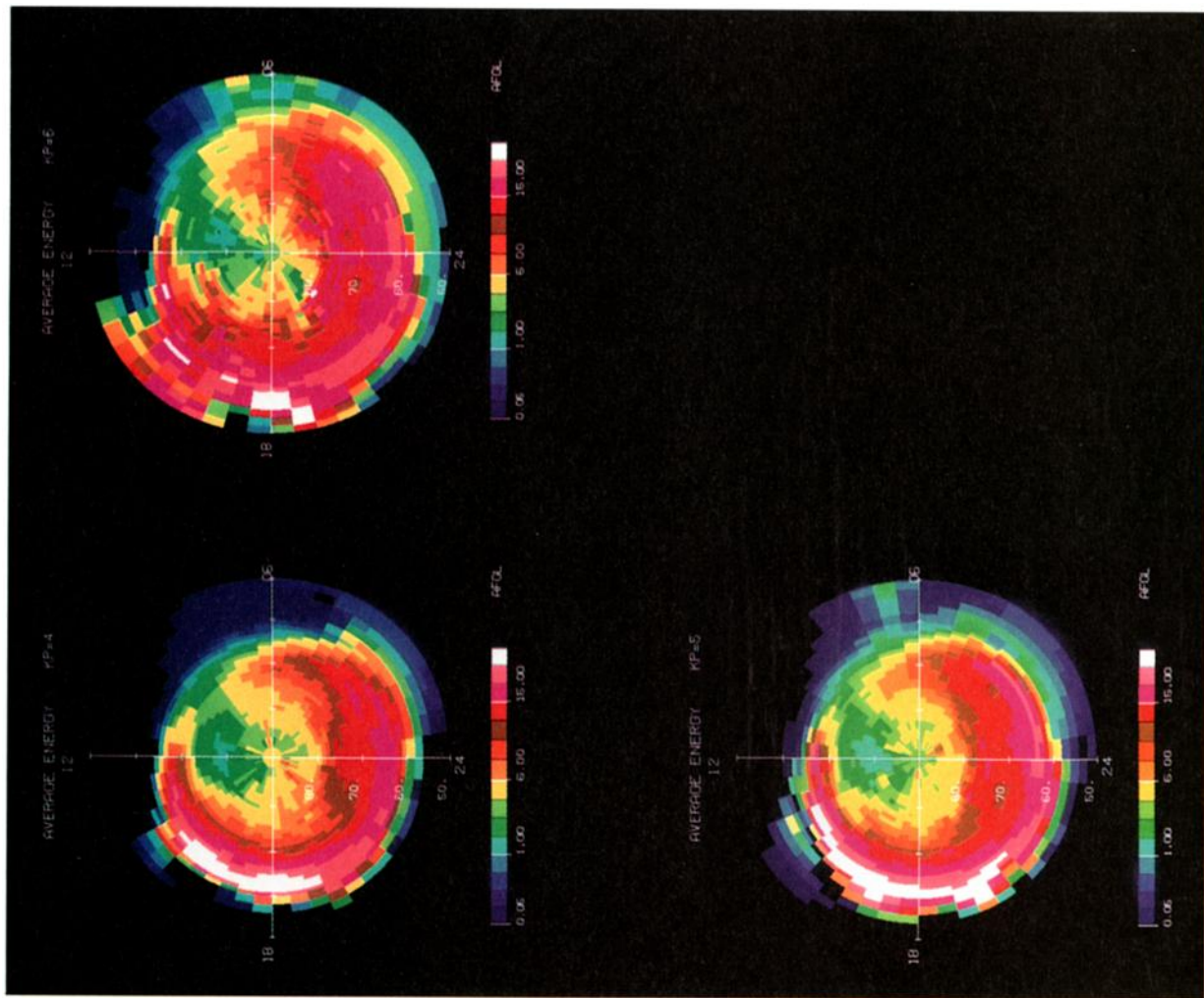


TABLE 3. Properties of the Nightside Energy Flux Maximum

K_p	MLAT, deg	MLT	JETOT, keV/cm ² s sr	EAVE, keV
0	67.5	2330–2400	3.93×10^7	9.69
1	66.5	2130–2200	5.52×10^7	13.6
2	65.5	2130–2200	9.68×10^7	17.3
3	65.5	2100–2130	1.23×10^8	19.5
4	63.5	2200–2230	1.35×10^8	21.0
5	62.5	2130–2200	1.67×10^8	17.2
$\geq 6-$	62.5	2300–2330	2.40×10^8	23.5

the mirror image sense (about the noon-midnight meridian) to the integral energy flux of electrons [Hardy *et al.*, 1985]. In Table 3 the characteristics and location of the integral energy flux maximum for each K_p are listed. As is the case for the midnight integral number flux maximum, the integral energy flux maximum moves equatorward by approximately 1° for each whole value increase in the value of K_p . The maximum is always found premidnight between 2100 and 2400 hours MLT. The value of the maximum varies by approximately a factor of 6 from 3.93×10^7 to 2.4×10^8 keV/cm² s sr (6.32×10^{-2} to 3.86×10^{-1} ergs/cm² s sr). The average energy varies from approximately 10 keV at $K_p = 0$ to 23.5 keV for $K_p \geq 6-$.

6. The integral energy flux has a single maximum in each one half hour MLT sector. The minimum value in MLT of this maximum occurs approximately prenoon. The location and ion characteristics at the minimum are listed in Table 4. While the level of integral energy flux on the nightside shows large increases with increasing K_p the level at the minimum increases by less than a factor of 2 from a value of 9.42×10^6 keV/cm² s sr for $K_p = 0$ to 1.71×10^7 keV/cm² s sr for the $K_p \geq 6-$ case (1.52×10^{-2} to 2.75×10^{-2} ergs/cm² s sr). The integral number flux varies over a similar small range. The minimum is found in the 0900–1100 MLT sector at all levels of activity. The ratio between the integral energy flux maximum in the premidnight sector and this minimum, increases from 4.17 for $K_p = 0$ to 14.0 for the $K_p \geq 6-$ case. Our observation of the softest ion spectra prenoon agrees with the results of Galperin *et al.* [1976].

7. The highest average energies are found on the eveningside of the oval with average energies exceeding 30 keV. The variation of the average energy with latitude is shown in more detail in Figure 3. In Figure 3 the values of the average energy along the dawn-dusk (Figure 3b) and the noon-midnight (Figure 3a) meridians are shown for the $K_p = 1, 3$, and 5 cases. For these plots the values were not smoothed, only averaged, over three adjacent MLT zones. Of the four local times shown, the highest average energies are along the dusk meridian. At dusk the average energy decreases

abruptly equatorward of the maximum, falling to values below 100 eV within 1° or 2° . At latitudes above the maximum the average energy decreases more slowly; falling monotonically to a value of 3–5 keV above 77° CGL in the polar cap region. On the dawn meridian the decrease in the average energy with latitude equatorward of the maximum is less sharp than along the dusk meridian. As on the eveningside, the decrease in the average energy poleward of the maximum is gradual; decreasing monotonically up to a latitude of 79° for $K_p = 1$ and $K_p = 3$ and to 85° for $K_p = 5$. At 0600 MLT the maximum value increases with increasing K_p from 6 to 9 keV as compared to an increase in the maxima from 17 to 32 keV at 1800 MLT. The fact that the ion spectrum is hardest on the eveningside of the oval and softens from the eveningside to morningside of the oval has been previously reported by Hultqvist [1974] and Galperin *et al.* [1976]. The average energy variation at midnight is much the same as that at dusk with an abrupt decrease equatorward and a gradual decrease poleward of the maximum average energy. The average energy maximum at noon is higher than that on the morningside at each of the three levels of activity, probably indicating that these ions penetrate to noon having drifted around the eveningside of the oval. For the two higher activity cases the peak average energy is greater at midnight than at noon.

8. There is a shift in MLT toward noon of the highest average energies as K_p increases from $K_p = 0$ to $K_p = 5$. In Table 5 the value of the maximum average energy and its location in MLT and latitude are listed for the seven K_p levels. Since the maximum value of the average energy fluctuates significantly as a function of MLT, the MLT for the maximum is defined as the center of the 4-hour MLT sector containing the highest values of the average energy.

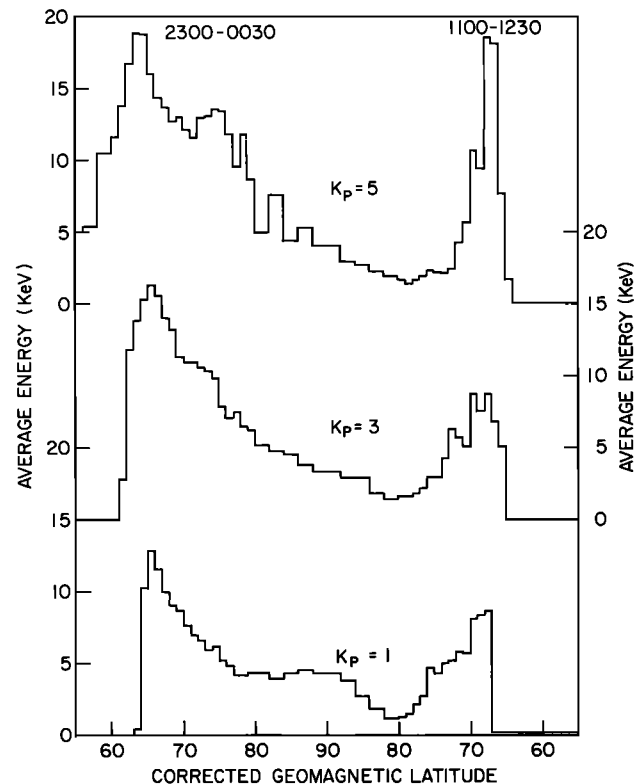


Fig. 3a. The average energy of the precipitating ions along the noon-midnight meridian for $K_p = 1, 3$, and 5.

TABLE 4. Properties of the Energy Flux Peak Minimum

K_p	MLAT, deg	MLT	JETOT, keV/cm ² s sr	JTOT, ions/cm ² s sr	EAVE, keV
0	78.5	1030–1100	9.43×10^6	4.16×10^6	2.27
1	77.5	1030–1100	1.23×10^7	4.83×10^6	2.54
2	76.5	1000–1030	1.45×10^7	4.96×10^6	2.92
3	75.5	1000–1030	1.60×10^7	5.33×10^6	3.00
4	74.5	0900–0930	1.70×10^7	5.26×10^6	3.23
5	72.5	0930–1000	1.67×10^7	5.66×10^6	2.95
$\geq 6-$	71.5	1000–1030	1.71×10^7	7.09×10^6	2.41

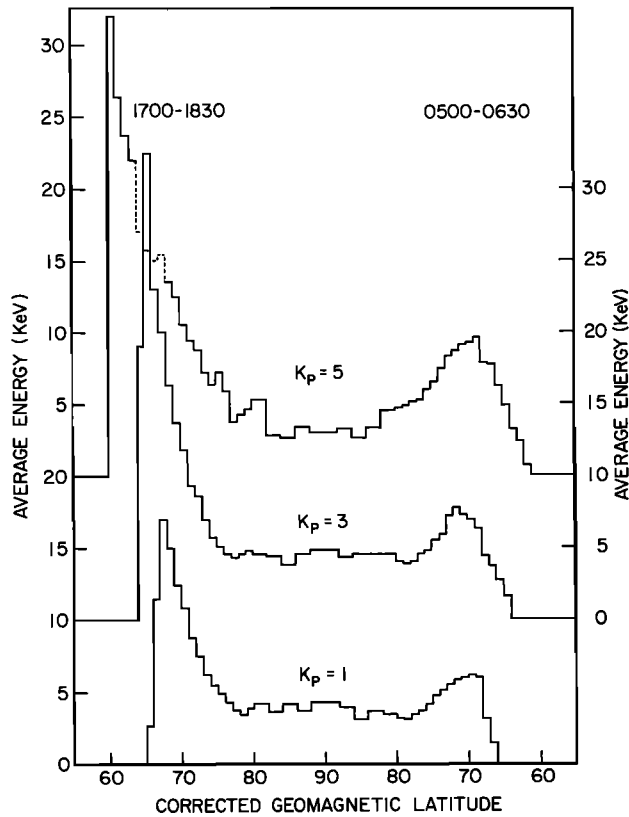


Fig. 3b. Same as Figure 3a but along the dawn-dusk meridian.

The maximum values and latitudes listed are averages over the 4-hour sector. For $K_p = 0$ the maximum is centered at approximately 2000 MLT and shifts to 1500 MLT for $K_p = 5$ case. For the $K_p \geq 6-$ case the location shifts back to 1800 MLT. This shift back to 1800 may be a statistical effect due to the relatively small sample available for the $K_p \geq 6-$ case. The latitude of the maximum moves equatorward with increasing K_p from 68.3° to 60.7° . The value of the maximum more than doubles from 14.6 to 36.8 keV from $K_p = 0$ to $K_p = 5$ and then decreases to 30.8 keV for the $K_p \geq 6-$ case. On average the value at the average energy maximum is approximately 1.75 times the maximum average energy at midnight.

9. There is a local minimum in the average energy at high latitudes on the dayside for all levels of activity. This cusp-associated feature is roughly circular and centered on local noon. In Figure 3a the minimum is seen located at 81° for $K_p = 1$ and 3 and at 78° for $K_p = 5$. In Table 6 the characteristics and latitude of the average energy minimum are listed. The value of the minimum is approximately

TABLE 5. Properties of the Average Energy Maximum

K_p	MLAT, deg	MLT	EAVE, keV
0	68.3	2000	14.6
1	66.9	1900	21.0
2	65.1	1800	27.4
3	64.8	1800	28.4
4	63.8	1700	30.2
5	63.7	1500	36.8
$\geq 6-$	60.7	1800	30.8

TABLE 6. Properties of the Average Energy Minimum

K_p	MLAT, deg	JTOT, ions/cm ² s sr	EAVE, keV
0	81.0	1.10×10^7	.974
1	81.0	5.67×10^6	1.12
2	81.0	4.32×10^6	1.13
3	81.0	2.83×10^6	1.46
4	79.5	4.36×10^6	1.80
5	78.5	2.65×10^6	1.41
$\geq 6-$	76.5	4.88×10^6	1.51

constant with values between 1 and 1.5 keV. The location of the minimum varies by only 5° . Comparing to Table 1, one notes that the ion cusp integral number flux maximum and average energy minimum are separated in latitude at all levels of activity. The separation increases with increasing activity from 1.5° for $K_p = 0$ to 7° for $K_p \geq 6-$. The values of the integral number flux at the average energy minimum are a factor of 2–5 less than that at the integral number flux maximum, but they are higher than the level of the integral number flux at the pole.

10. The equatorward boundary of auroral ion precipitation, defined as the minimum latitude at which the average energies exceed 1 keV, is roughly circular. In Figure 4 the location of this boundary is plotted for the $K_p = 2$ case along with the equatorward boundary for electron precipitation using the equations of Gussenhoven et al. [1983]. We fit the ion boundary to a circle with a radius of 26.0° and a center offset from the geomagnetic pole by 3.5° slightly toward the premidnight region. The electron boundary by contrast has a radius of approximately 24.5° and is offset 3.5° toward the

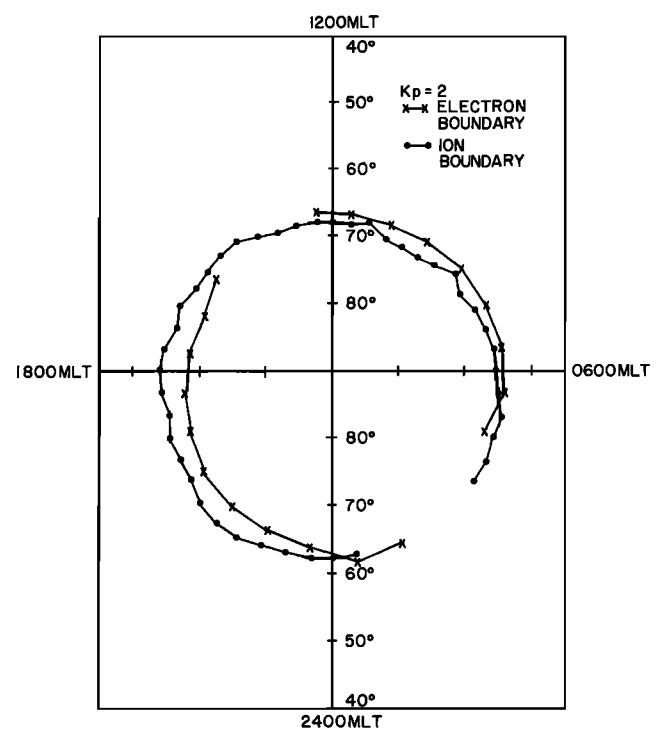


Fig. 4. The equatorward boundary of ion precipitation, defined as the point where the average energy exceeds 1 keV and the equatorward boundary of electron precipitation as defined by Gussenhoven et al. [1983]; both for $K_p = 2$ and both plotted in a corrected geomagnetic latitude-magnetic local time coordinate system.

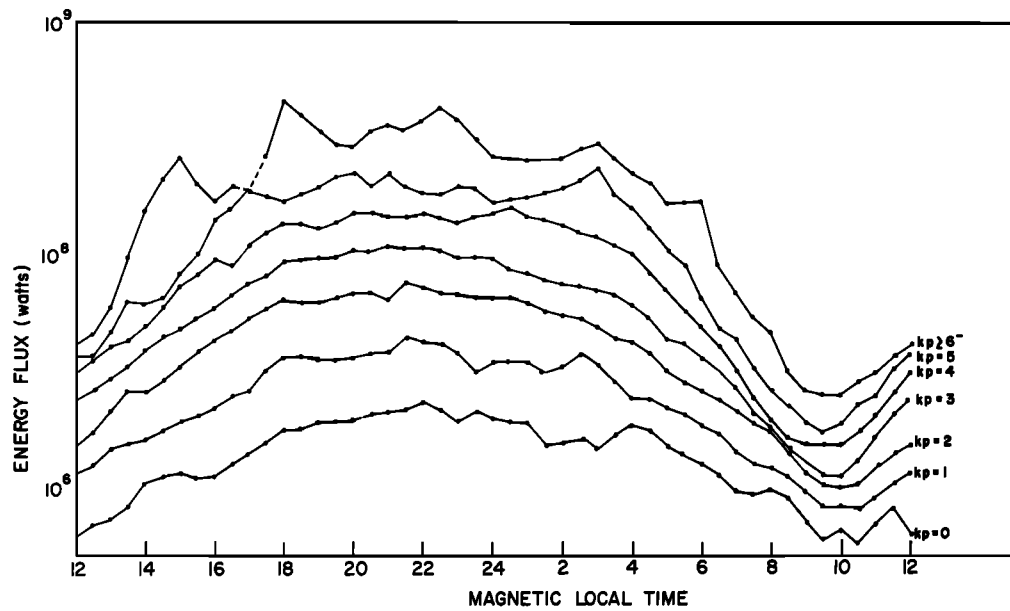


Fig. 5. The energy flux from the ions integrated over latitude and plotted as a function of MLT for each of the seven levels of activity.

postmidnight region. On the eveningside the boundary of the ion precipitation extends as much as 4° equatorward of the electron precipitation. On the morningside by contrast the electrons extend slightly equatorward of the ion boundary: of the order of 1° – 2° . The two boundaries are seen to coincide approximately in the postnoon and postmidnight regions. This same approximate behavior is repeated at each level of activity. This behavior agrees with the determination of the ion boundaries and their separation given by *Gussenhoven et al.* [1987]. This behavior in the boundaries was also seen in the average electron and ion maps of *Reidler* [1972] and *Reidler and Borg* [1972].

In Figures 5 and 6 the total ion energy flux and the total ion number flux integrated over each half hour in MLT and over latitudes from 50° to 90° CGL are plotted as a function of magnetic local time for each of the seven levels of Kp . For both quantities, except for the $Kp = 0$ trace and the highest-activity trace, the curve for each level of activity is separated from all others at all MLTs. For the integral number flux there is no consistent maximum in magnetic local time at any of the seven levels of activity. The level of number flux varies by no more than a factor of 2 at every level of activity with the values falling between 10^{22} and 10^{23} ions/s sr. The contribution to these values from the extrap-

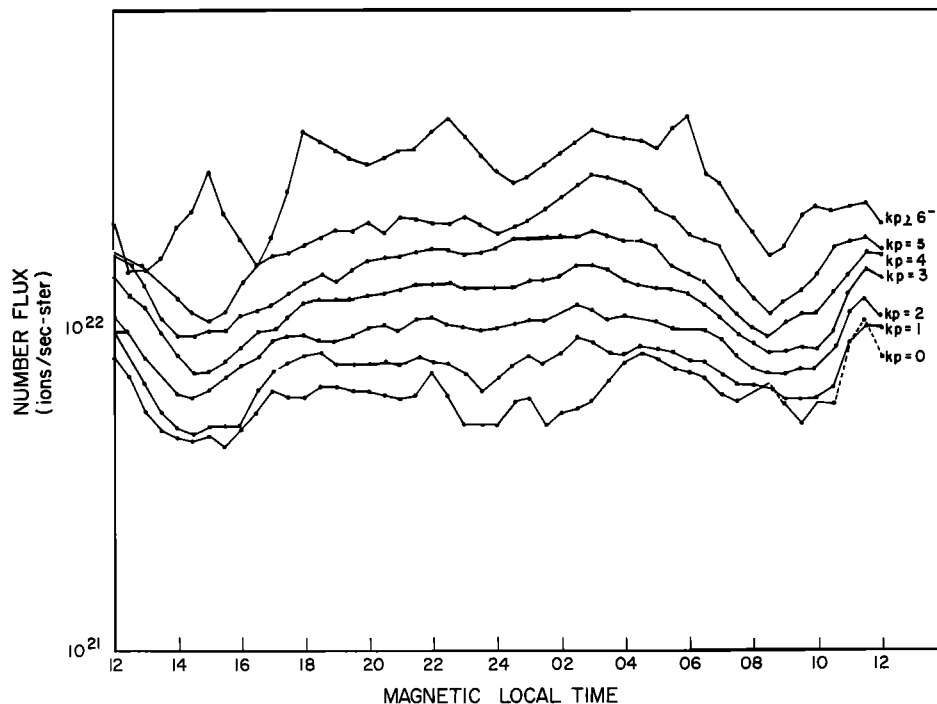


Fig. 6. Same as Figure 5 but for the number flux integrated over latitude.

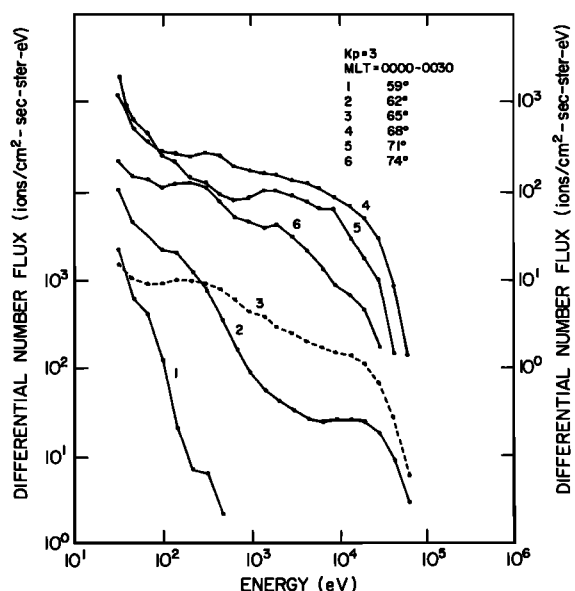


Fig. 7a. The logarithm of the average differential number flux spectra at midnight MLT for $K_p = 3$ plotted versus the logarithm of energy. The spectra are plotted at 3° intervals. The scale for the first three spectra is on the left-hand side and the scale for the remaining spectra is on the right-hand side.

olation typically does not exceed 25%. On the morningside of the oval the contributions are generally in the range from 5 to 10%, and on the eveningside they range from 10 to 25%. The contribution from the extrapolation increases gradually with K_p .

As for the integral number flux, the integral energy flux smoothly varies in MLT. The maximum flux is observed premidnight. The minimum is seen consistently at approximately 1000 MLT in agreement with the polar spectrograms and the minima listed in Table 4. The ratio of the maximum energy flux to the minimum energy flux increases with increasing activity. For the seven levels of activity the ratios are 3.8, 5.3, 7.1, 8.9, 9.1, 10.9, and 11.4. The increase in the ratio reflects an increase in the level of ions injected or transported into the nightside with increasing activity but the

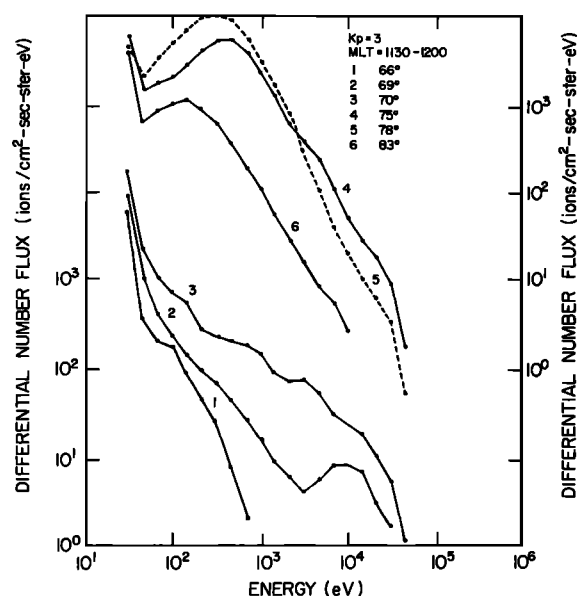


Fig. 7c. Same as Figure 7a but for 1130–1200 MLT.

inability of these ions to be either transported and/or precipitated in the prenoon region. This behavior is similar to that seen in our earlier work for the hot electrons at noon and postnoon [Hardy *et al.*, 1985].

As for the number flux, the contribution of the extrapolation to these values increases with increasing K_p . The largest contributions are on the evening and midnight sectors where the percentage increase varies from less than 10% for $K_p = 1$ to approximately a factor of 2 for $K_p \geq 6$. At 0600 the contribution is less than 15% at all K_p . At noon the contribution varies from less than 10% at $K_p = 1$ to approximately 50% for $K_p \geq 6$.

Ion average spectra. Examples of the average ion spectra and their variation in magnetic local time and magnetic latitude are shown in Figure 7. In Figure 7, average differential number flux spectra have been plotted for $K_p = 3$ at 3° CGL intervals for the midnight (Figure 7a), dawn (Figure 7b), noon (Figure 7c), dusk meridians (Figure 7d). Although

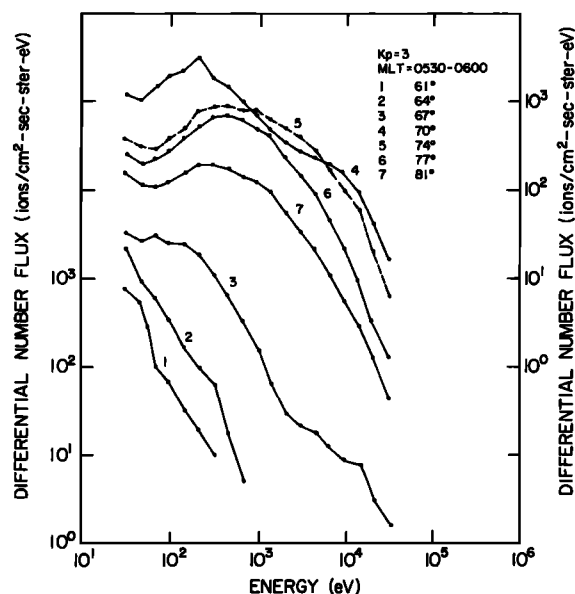


Fig. 7b. Same as figure 7a but for 0530–0600 MLT.

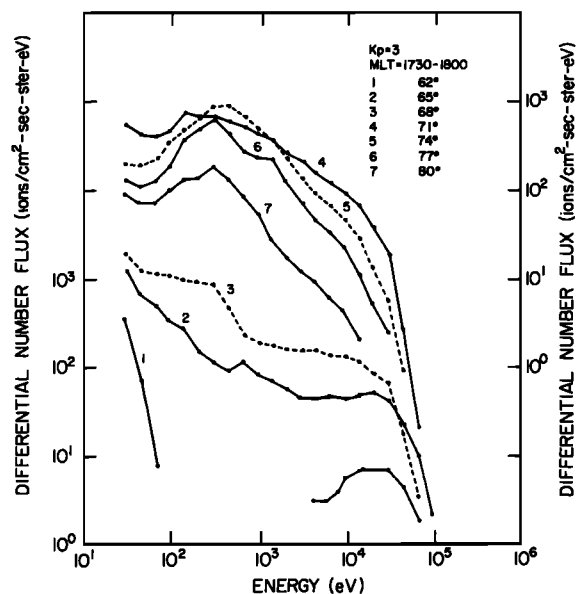


Fig. 7d. Same as Figure 7a but for 1730–1800 MLT.

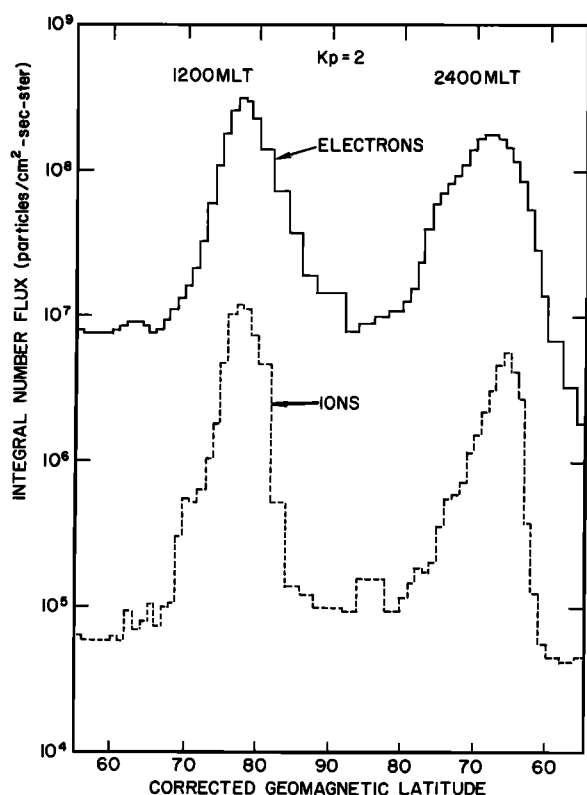


Fig. 8a. The average integral number flux for electrons and ions along the noon-midnight meridian for $K_p = 2$.

only spectra from the $K_p = 3$ case are shown, they are broadly representative of the spectral shapes seen at the other activities. Changing the K_p level tends to change the overall intensity of the spectrum more than the spectrum's shape. Figure 7 illustrates the following:

1. At midnight, noon, and 1800 a very hot, weak spectrum is seen at or near the equatorward edge of the ion precipitation. At high energies this spectrum is approximately Maxwellian. It is hottest at 1800 MLT with a peak in the differential number flux spectrum at approximately 20 keV. At midnight and noon the peak is closer to 10 keV.

2. Up to 69°, the spectrum at 1800 is monotonically decreasing from 30 eV to approximately 1 keV. From approximately 1 to 10 keV the spectrum is flat and then falls off for higher energies. Such two-component spectra have been reported by Galperin *et al.* [1976]. The softer spectra, at latitudes above 69°, have a clear peak at values between 200 and 500 eV. At midnight the lowest-latitude spectrum is very cold and monotonically decreasing. The midnight spectra first harden with increasing latitude and then soften. The spectra soften more slowly with latitude than at 1800 and show no development of a peak below 1 keV for any latitudes outside of the polar cap region. Instead, at higher latitudes the spectrum tends to be monotonically decreasing in intensity with increasing energy over the entire energy range.

3. At 0600, as at midnight, a very cold ion spectrum, monotonically decreasing in intensity with energy, is found at the lowest oval latitudes. The spectrum initially hardens with increasing latitude from the boundary. At all latitudes above 67° the differential number flux spectra have a clear peak in the energy range from 200 to 500 eV. The shape of the spectra at these latitudes is basically the same as seen

above 74° CGL at 1800 MLT. The most intense spectrum is seen at approximately 70° CGL with a peak at approximately 200 eV. At latitudes above 70° CGL the spectrum first hardens slightly and then decreases in intensity and softens with increasing latitude. At all MLTs the spectra toward the poleward edge of the oval, although often peaked at a few hundred electron volts, generally have a significant high-energy tail extending out to at least 30 keV. The presence of such high-energy tails results in the relatively high average energies seen in Figure 3 and in the spectrograms.

4. At noon, as at 0600, the lowest-latitude spectrum is very cool and monotonically decreasing in intensity with increasing energy. At 69° CGL a hot, but very weak, spectrum is seen with a peak at 10 keV. Poleward of this the spectrum softens and displays a peak at energies of 200–500 eV. The spectrum with the maximum integral number flux is at a latitude of approximately 75° CGL. The peak intensity for the spectrum at 75° exceeds that for the peaked spectra at 1800 and 0600 MLT by factors of 10 and 3.3, respectively. The peaked spectra at noon have less intense high-energy tails than the peaked spectra at 0600 and 1800 MLT. The ion spectrum at 75° agrees in shape and intensity with other spectra measured in the cusp.

Ion and Electron Comparisons

The comparison of the average characteristics of the electron and ion precipitation is limited to the $K_p = 2$ and $K_p = 3$ cases. At all other activities much the same relative behavior is found. In Figures 8, 9, and 10 the integral number flux, integral energy flux, and average energy for electrons and ions are plotted. The ion data are from the present data set and the electron data are taken from our previous work [Hardy *et al.*, 1985]. Figures 8a, 9a, and 10a are the profiles

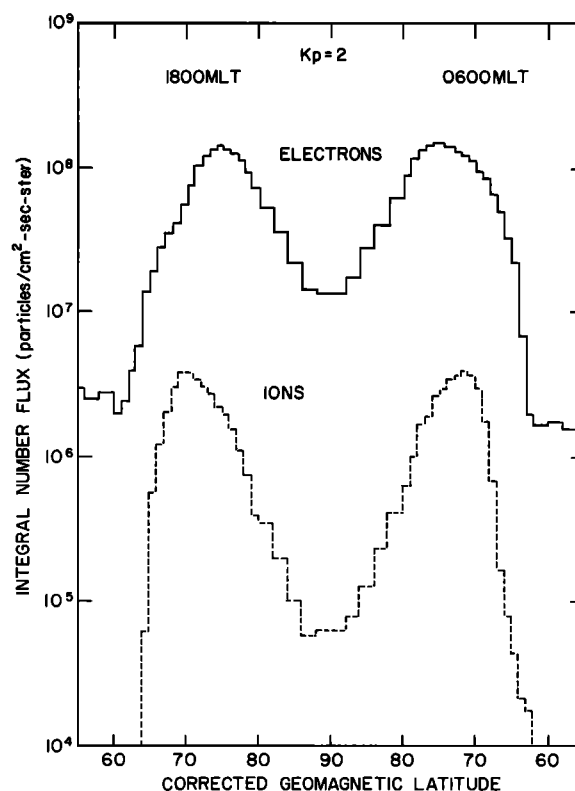


Fig. 8b. Same as Figure 8a but for the dawn-dusk meridian.

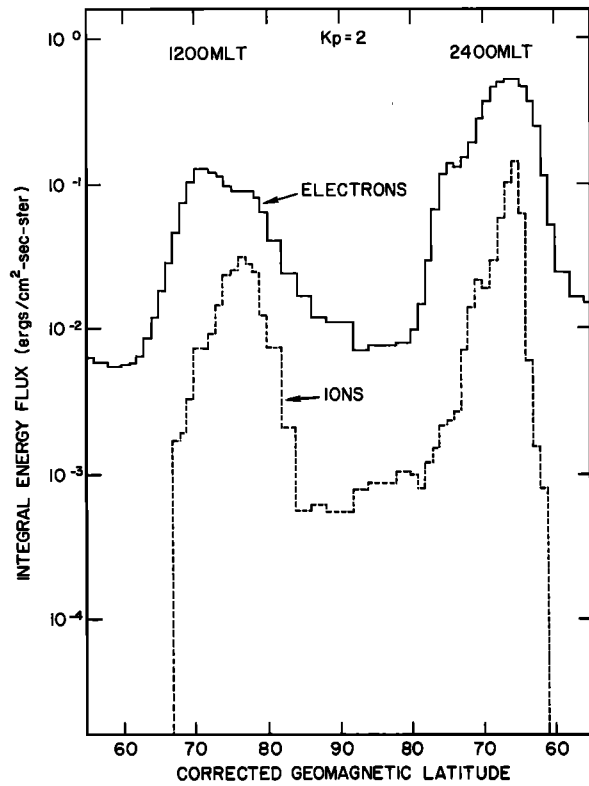


Fig. 9a. The average integral energy flux for electrons and ions along the noon-midnight meridian for $K_p = 2$.

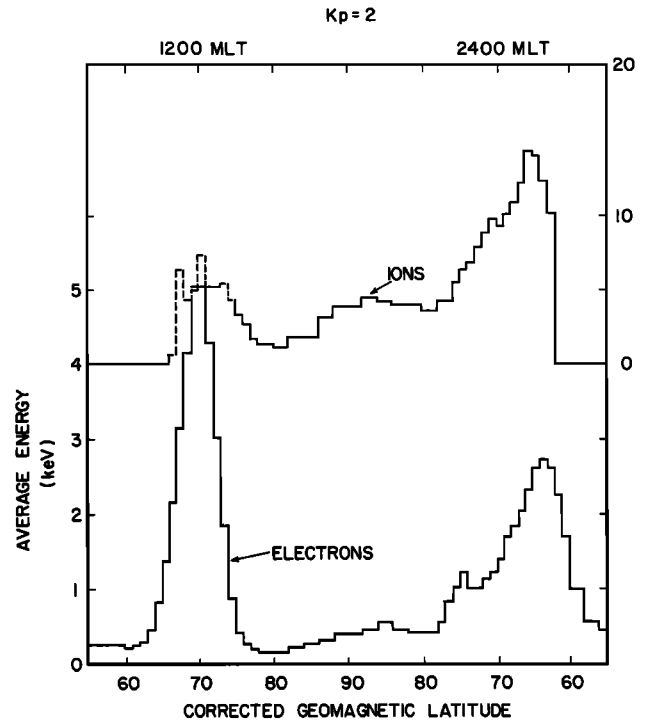


Fig. 10a. The average energy for electrons and ions along the noon-midnight meridian for $K_p = 2$. The scale for the ions is to the right of the figure that for electrons is to the left.

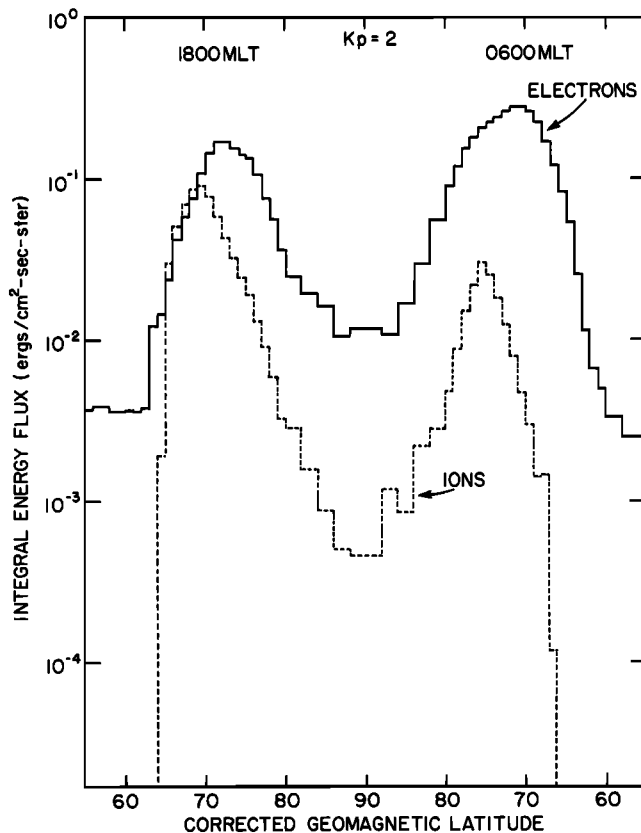


Fig. 9b. Same as Figure 9a but along the dawn-dusk meridian.

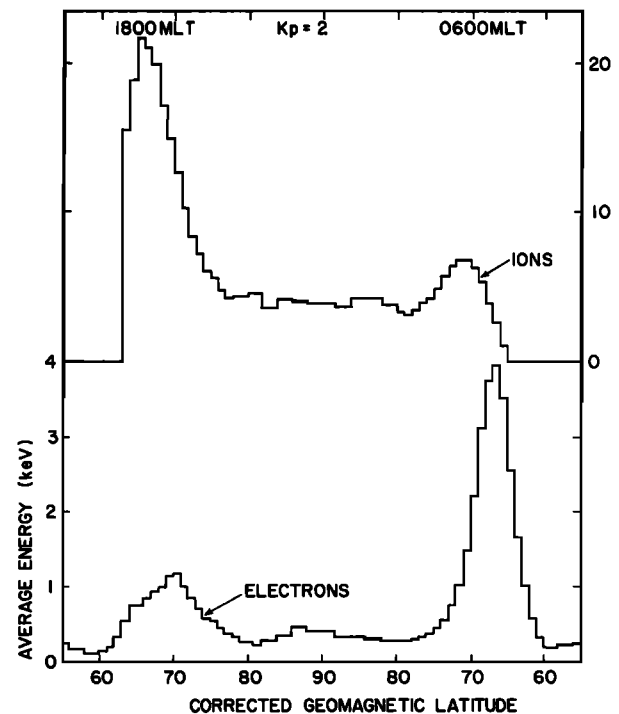


Fig. 10b. Same as Figure 10a but along the dawn-dusk meridian.

along the noon-midnight meridian, and Figures 8b, 9b, and 10b are the profiles along the dawn-dusk meridian. Figures 8–10 illustrate the following:

1. The integral number flux of the ions is always small compared to that of the electrons; generally smaller by from 1–2 orders of magnitude as has been reported from the study of individual passes [Lui *et al.*, 1977]. Even at noon where

we take the peak ion integral number flux to be at the center of the cusp the ion integral number flux is approximately 10^7 ions/cm² s sr as compared to an electron integral number flux of 3×10^8 electrons/cm² s sr. Since the ion integral number flux is everywhere always such a small fraction of the electron integral number flux, the current carried by the precipitating ions in this energy range does not, on average, contribute significantly to the region 1 and 2 currents.

2. The energy flux carried by the ions can be a significant fraction of the energy flux carried by the electrons. On the eveningside, near the electron equatorward boundary, the ion integral energy flux equals or somewhat exceeds the electron integral energy flux. This behavior is seen at all levels of activity for several hours in MLT centered at 1800. In no other local time sector or latitude range does the average ion energy flux exceed the average electron energy flux. At midnight, by comparison, at the peak of the ion integral energy flux, the ratio of the ion to electron energy flux is 0.28. On the morningside at 0600 the contribution of the ions is smallest with a ratio of 0.14 at the peak. At noon, at the center of the cusp the ratio is 0.43. Over all seven activity levels the ratio of the ion to electron energy flux in the center of the cusp is reasonably constant varying from 0.33 for $K_p = 1$ to 0.47 for $K_p = 5$ with an average over the seven levels of 0.44.

3. In the dawn-dusk meridian the pattern for the ion energy flux is clearly displaced equatorward of that for electrons on the eveningside and poleward on the morningside. This is in the same sense as the reported displacement in the peak electron and ion associated optical emissions [Wiens, 1968] and in the displacement of peak intensities in the statistical maps of ion and electron precipitation of Reidler [1972] and Reidler and Borg [1972]. The relationship between the location of the peak in the ion and electron energy flux is shown in Figure 11 for the $K_p = 3$ case. In

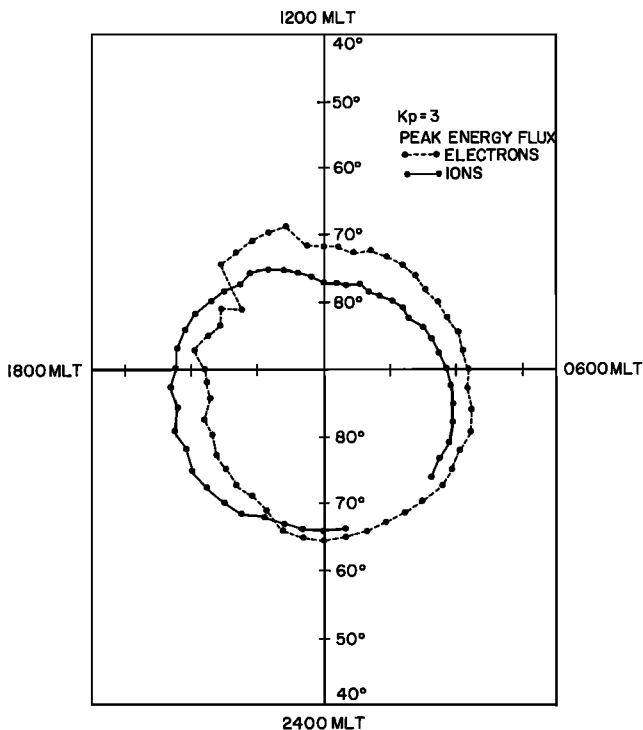


Fig. 11. The location of the peak electron and ion energy flux in latitude plotted as a function of magnetic local time.

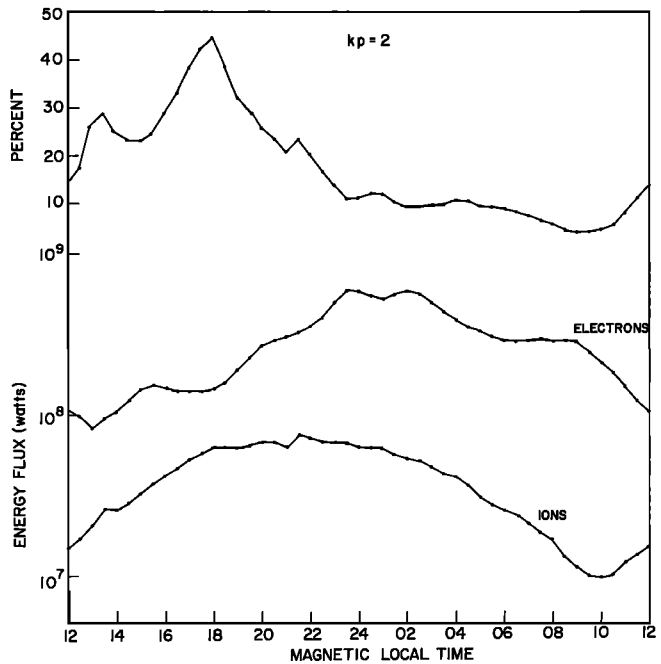


Fig. 12. The energy flux from electrons and ions integrated over latitude is plotted as a function of magnetic local time. The ratio of the ions energy flux to electron energy flux is plotted at the top of the figure versus magnetic local time.

Figure 11 the geomagnetic latitude of the peak for the electron and ion integral energy flux is plotted as a function of magnetic local time in a polar projection. On the eveningside, between 1400 hours MLT and midnight, the ion peak is 2° – 4° equatorward of the electron peak. At midnight the location of the two peaks nearly coincide. On the morningside the behavior is reversed with the electron peak equatorward of the ion peak and with the difference in latitude of the two peaks increasing with magnetic local time toward noon. At noon the difference is greater than 6° . For the electrons there is an abrupt shift in the location of the peak toward higher latitudes at approximately 1400 hours MLT. For the ions the poleward shift is more gradual. For the ions the shift is caused by a decrease with MLT toward noon in the precipitation level of hot ions drifting around on the eveningside of the oval such that the peak becomes defined by the ion precipitation in the cleft or cusp. Similarly, for the electrons, the abrupt change in the latitude of the peak is the result of the decrease past noon MLT in the level of precipitation of hot electrons drifting around on the morningside of the oval such that the peak becomes defined by the precipitation in the cusp or cleft or by colder electrons drifting around on the eveningside of the oval.

4. In the dawn-dusk meridian the highest ion average energies are seen at dusk, and the highest electron average energies are seen at dawn. For electrons (ions) the ratio of the peak average energy at dawn (dusk) to that at dusk (dawn) is approximately 3.4 (3.2). At dusk the ion average energy peaks equatorward of the electron average energy, and at dawn the behavior is reversed. Along the noon-midnight meridian the peak average energy for the ions is at midnight and that for electrons is at noon. The regions of hot electrons and ions at noon roughly coincide, while at midnight the electron average energy peaks equatorward of the ion average energy.

TABLE 7. Total Hemispheric Contribution of Electron and Ions to the Number and Energy Flux

Kp	Number Flux, ions/s sr	Number Flux, electrons/s sr	Ion Energy Flux, W	Electron Energy Flux, W
0	$3.62 (3.46) \times 10^{23}$	1.53×10^{25}	$7.26 (6.74) \times 10^8$	4.68×10^9
1	$3.80 (3.72) \times 10^{23}$	1.85×10^{25}	$1.38 (1.26) \times 10^9$	8.14×10^9
2	$4.54 (4.25) \times 10^{23}$	2.35×10^{25}	$2.03 (1.61) \times 10^9$	1.49×10^{10}
3	$5.53 (5.05) \times 10^{23}$	3.08×10^{25}	$2.92 (2.06) \times 10^9$	2.59×10^{10}
4	$7.10 (6.00) \times 10^{23}$	4.05×10^{25}	$4.10 (2.43) \times 10^9$	3.36×10^{10}
5	$8.74 (7.10) \times 10^{23}$	5.83×10^{25}	$5.47 (2.75) \times 10^9$	4.32×10^{10}
$\geq 6-$	$1.50 (1.29) \times 10^{24}$	1.02×10^{26}	$1.00 (0.569) \times 10^{10}$	7.76×10^{10}

Extrapolated values are without parentheses, and unextrapolated values are in parentheses.

A more quantitative display of the relative contribution of the electrons and ions to the energy flux in the oval is shown in Figure 12. In Figure 12 the energy flux for the electrons and ions, for $Kp = 2$, integrated over latitude, is plotted as a function of MLT. In addition, the ratio of the ion to electron energy flux is shown. Across the eveningside the ratio is seen to generally exceed 20%, reaching a maximum of 45% at 1800 MLT. From 1800 MLT to approximately midnight the ratio is primarily decreasing. On the morningside the ratio is approximately 10% from midnight to 0600 and then decreases to a minimum value of 4% at approximately 1000 MLT. This same approximate behavior is maintained at other levels of activity.

In Figures 13 and 14 the total hemispheric energy and particle inputs for both electrons and ions are plotted as a function of magnetic activity as well as the ratio of the ion to electron levels. Both the extrapolated and unextrapolated values for the ions are listed in Table 7. The unextrapolated values are shown in parentheses. The results for the number flux reiterate the points made above. The ion number flux varies from 3.6×10^{23} ions/s sr to 1.5×10^{24} ions/s sr as compared to a range for the electrons of 1.5×10^{25} electron/s sr to 1.0×10^{26} electrons/s sr. The ratio for the number flux is largest at $Kp = 0$ with a value of 0.024 and decreases as a function of increasing activity to a value of 0.015 for $Kp \geq 6-$. The total increase in the global number flux of ions is only a factor of 4.1, while for the electrons the increase is a factor of 6.7. The energy flux for the ions ranges from 7.3×10^8 W to 1.0×10^{10} W and the electrons from 4.7×10^9 W to 7.8×10^{10} W. The total energy flux increase is a factor of 13.8 for the ions and a factor of 16.6 for the electrons. Most significantly the ratio of the ion to electron energy flux ranges from 0.11 to 0.17 over all levels of activity. The largest ratios are seen for $Kp = 0$ and 1 but even for the $Kp \geq 6-$ case the ratio is 0.13. Averaging over all activities the ratio is 0.136. The energy flux of the ions on a global scale is therefore significant relative to the energy flux from the electrons. The extrapolation increases the global number flux by a minimum of 2.2% for $Kp = 1$ and a maximum of 23.1% for $Kp = 5$. For the energy flux the increase varies from 7.7% for $Kp = 0$ to 99% for $Kp = 5$. As noted in the appendix, the procedure used to extrapolate the spectra in energy most likely underestimates the energy above 30 keV such that the ratios should be taken as lower bounds.

5. DISCUSSION

We have shown in the preceding section that, on average, the precipitating ions in the auroral zone occur in a regular

pattern that varies systematically in magnetic local time and geomagnetic latitude and with activity as measured by Kp . In this section we will discuss what can be inferred from the ion patterns about the processes in the regions of the magnetosphere connecting to the auroral zone, along with the degree to which the average behavior of the ions agrees with previous measurements made in the auroral zone. Specifically, we will discuss the degree to which the variation in ion characteristics, at least over some range of latitudes, MLTs, and activities, is consistent with a simple physical model of adiabatic compression and the creation of Alfvén layers as the ions are transported within the magnetosphere. The chief motivation for this is to determine the degree to which such adiabatic effects, reported by earlier researchers in case studies, are repeated in the average study [Hultqvist, 1974; Galperin *et al.*, 1976, 1978; Hultqvist, 1979, and references therein; Sauvaud *et al.*, 1981]. In addition, we will consider the relationship of the precipitating ions to the various boundary regions in the magnetosphere, the implications of the variation of the average energy and integral number flux in the cusp region, and the degree to which the pattern of ion precipitation agrees with models for the motion of ions in the magnetosphere.

We first consider the role of adiabatic compression and Alfvén layers in determining the global pattern of auroral ion precipitation. For plasma in a dipole field, in the small displacement limit, the average energy of the ions $EAVE(\alpha)$ at an equatorial pitch angle α varies with L shell as

$$EAVE(\alpha) = EAVE_0(\alpha)(L_0/L)^{v_T(\alpha)} \quad (5)$$

where $EAVE_0(\alpha)$ is the initial average energy at $L = L_0$ and $v_T(\alpha) = 6A(\alpha)/T(\alpha)$ with $T(\alpha)$ and $A(\alpha)$ integrals related to the bounce period and the second adiabatic invariant [Cowley and Ashour Abdalla, 1975; Kivelson and Southwood, 1975]. This relationship is exact for 0° and 90° pitch angle particles where

$$EAVE(0^\circ) = EAVE_0(0^\circ)(L_0/L)^2 \quad (6)$$

$$EAVE(90^\circ) = EAVE_0(90^\circ)(L_0/L)^3 \quad (7)$$

The two cases bound the value of the exponent which, for arbitrary initial α , varies with L shell.

The above relationship is derived for a dipole magnetic field with a uniform and azimuthally symmetric source at some fixed radial distance. Since the conditions in the magnetosphere strongly deviate from this, the actual behavior is expected to agree only qualitatively with the equations. For example, there is ample evidence in the literature that the impulsive injection of plasma into the inner magneto-

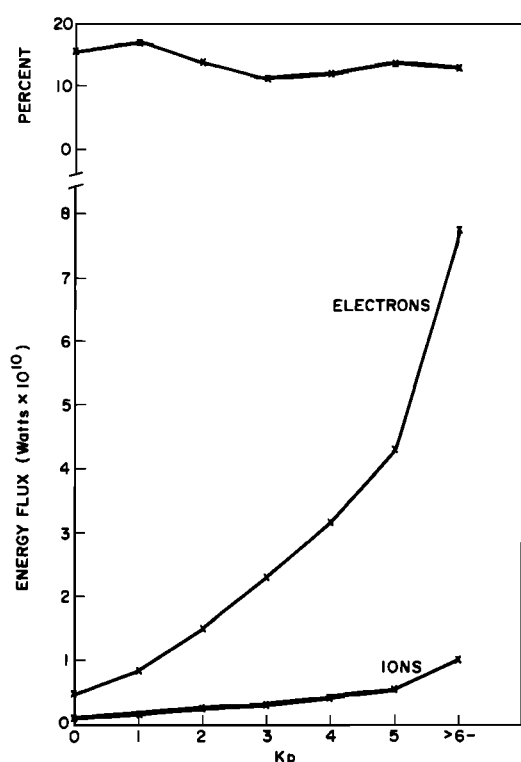


Fig. 13. The hemispheric precipitating electrons and ions energy flux plotted as a function of Kp . The top of the figure gives the ratio of the ion to electron energy input.

sphere plays a key role in determining plasma characteristics in that region and in particular in establishing the earthward boundary to which that plasma penetrates [McIlwain, 1974; Konradi *et al.*, 1975; Mauk and Meng, 1983a, b, 1986]. There is a body of research involved with the implications of the characteristics of the boundary. There has been debate as to whether the characteristics of the boundary are the product of creation of steady state Alfvén layers [Kivelson, 1975], Alfvén layers with a time varying electric field [Kaye and Kivelson, 1979; Chiu and Kishi, 1984], particle injection to an injection boundary (references as above) or earthward moving injection fronts [Moore *et al.*, 1981]. In this analysis we are less interested in the boundary than in looking at the ion characteristics as a function of latitude well removed from the boundary. As pointed out above, previous research has shown [Galperin *et al.*, 1976, 1978; Hultqvist, 1979; Sauvaud *et al.*, 1981] that over a range of latitudes, for individual passes in the evening and morning sectors and during quiet times, the average energy of the precipitating ions varies with L shell approximately as would be expected if the ions were undergoing quasi-static, adiabatic compression.

To investigate the degree to which the average patterns are consistent with a simple picture of adiabatic compression, the average energy of the ions was plotted as a function of $1/L^2$ at 1800, midnight, and 0600 MLT (Figures 15, 16, and 17). The L shell, in this case, is calculated for a dipole magnetic field. For 1800 and 0600 MLT the data are plotted for the six Kp levels from $Kp = 0$ to $Kp = 5$. The $Kp \geq 6$ case was excluded because the statistical sample was too small to give a clear result. At midnight only the $Kp = 0$ to $Kp = 3$ cases are shown. At midnight and 0600 the average energy at each Kp level is normalized to the peak average

energy at that Kp level. A similar trend was obtained when the data were plotted versus $1/L^3$.

The variation in the average energy along the dusk meridian appears to be reasonably consistent with adiabatic compression. At this local time the average energy variation with $1/L^2$ follows the same, approximately linear trend at each activity level down to a minimum L shell, below which the average energy decreases rapidly. This minimum L shell decreases with increasing activity. We attribute this decrease in L shell to an increase in the average cross-tail electric field with increasing activity such that the plasma is transported to smaller radial distances, as is seen in the motion of the equatorward boundary of auroral precipitation [Gussenhoven *et al.*, 1981, 1982, 1983]. The average energy decrease may reflect the action of Alfvén layers. As pointed out by Cowley and Ashour-Abdalla [1975], the average energy of plasma convected radially inward in a dipole magnetic field will increase linearly with $1/L$ to a power as long as the average energy of the ions at a particular L shell is much less than the cutoff energy for the Alfvén layer at that L shell. As the cutoff energy for the Alfvén layer becomes comparable, the average energy decreases since a significant portion of the particle distribution is excluded from that L shell. The falloff in average energy need not be the result of Alfvén layers. The decrease could result, just as well, from dispersions developed as the results of the motion of particles from the injection boundary [Mauk and Meng, 1986] or could be in part an artifact of the averaging process.

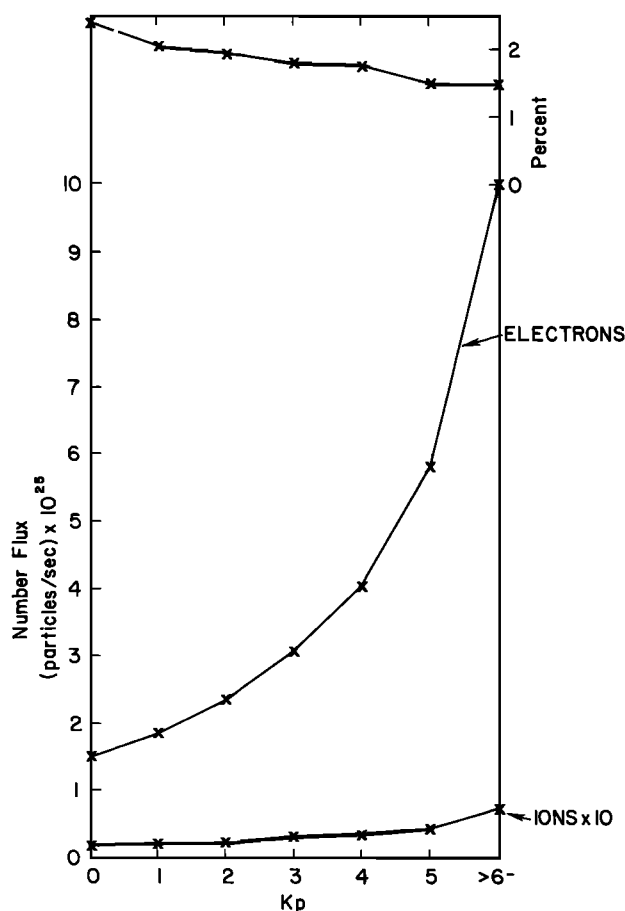


Fig. 14. Same as Figure 13 but for the hemispheric number flux.

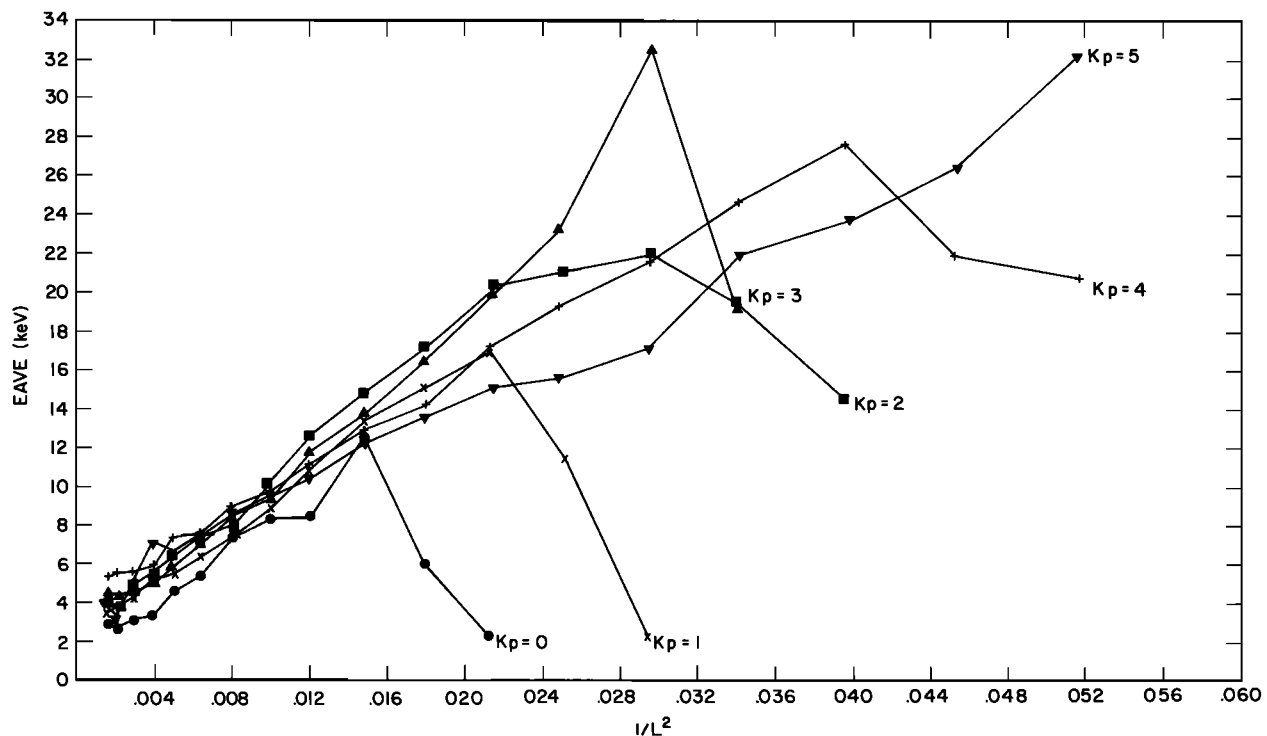


Fig. 15. The average energy at the dusk meridian plotted versus $1/L^2$ for each of the seven activity levels. The L shell was calculated using a dipole field model.

At 0600 the ion average energy varies linearly with $1/L^2$ at all Kp for L shells at latitudes above approximately 71° CGL. Below 71° CGL the average energy at each Kp level decreases. At these lower latitudes the average energy at a fixed L shell generally increases with increasing Kp as does the minimum L shell to which the hotter ions are observed. Sauvaud *et al.* [1981] showed much the same behavior for individual passes from the ARCAD experiment during quiet geomagnetic conditions. Our results show that this behavior exists at all levels of activity. A reason for the difference from the eveningside behavior may be the presence of a relatively low-energy ionospheric contribution to the precipitation at lower latitudes [Newell and Meng, 1986].

At midnight the average energy variation with L shell displays a much less consistent behavior with activity than that seen at dawn and dusk. For the $Kp = 0$ case the average

energy is roughly constant for latitudes above approximately 70° CGL then rises sharply for lower latitudes. For $Kp = 1, 2$, and 3 the L shell variation is similar to, but not so clear as, that on the dusk meridian, while for higher Kp (not shown) the average energy is high and approximately constant over much of the auroral region. There are two possible sources for the breakdown in the relationship with increasing Kp . First, in the early stages of a substorm the magnetic field in the midnight sector is known to become more taillike due to an intensification and inward motion of the neutral sheet current. At substorm onset the field abruptly returns to a more dipole configuration [Russell and McPherron, 1973; Moore *et al.*, 1981]. For such a situation the use of a dipole magnetic field model to map from the auroral zone into the tail is incorrect. Second, during a substorm there is a sudden injection of plasma into the inner magnetosphere [McIlwain,

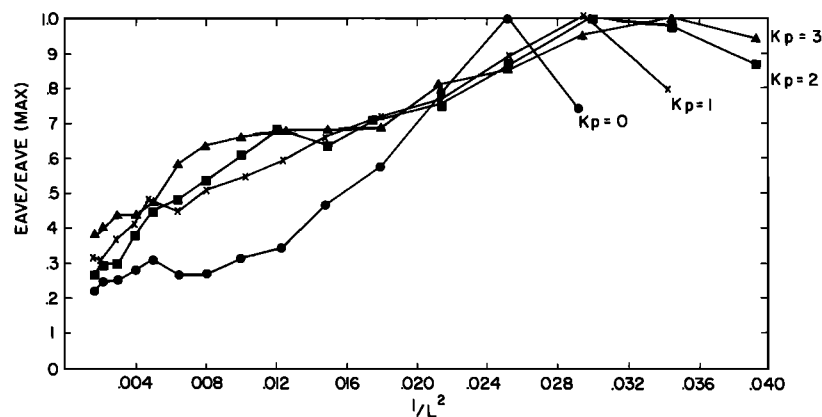


Fig. 16. The average energy at the midnight meridian plotted versus $1/L^2$ for the $Kp = 0, 1, 2$, and 3 cases. The average energy in each curve has been normalized to the maximum average energy observed at midnight for that activity level.

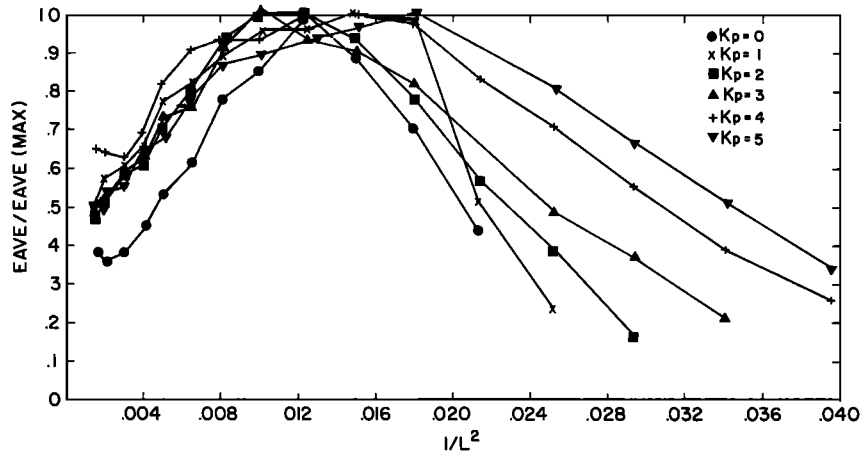


Fig. 17. The average energy at the dawn meridian plotted versus $1/L^2$ for six levels of activity. Each curve has been normalized to the maximum average energy observed at the dawn meridian for that activity level.

1974]. The magnetic and electric fields in this process change abruptly such that the adiabatic invariants may be violated and variations based upon their conservation will no longer hold. The fact that such a breakdown in the variation is not seen at the dawn and dusk flanks suggests that such non-adiabatic substorm effects are stronger in the midnight sector. The relatively wide L shell range where the average energy is constant and relatively low for $Kp = 0$ may reflect a large expansion in the size of the boundary plasma sheet for low activity.

The poleward portion of the region of enhanced ion precipitation must map to the cusp and the various boundary regions in the magnetosphere: the low-latitude boundary layer, the plasma mantle, and the boundary plasma sheet [Vasyliunas, 1979]. We have shown previously [Hardy et al., 1985] that the electrons from the cusp and boundary regions are distinguishable from those of the central plasma sheet by a sharp difference in the electron average energy (one to a few hundred eV in the former one to several keV in the latter). The ion precipitation, by contrast, shows no such clear average energy separation except in the region near the average energy minimum at noon. We have shown, however, that at noon, dawn, and dusk a separation can be made, based upon spectral shape, between a region at low latitudes populated by hot, relatively thermal ions and a region at higher latitudes populated by softer ions with a spectrum peaked at a few hundred eV. The similarity in shape of these peaked spectra to the shape of spectra in the magnetosheath suggests that these peaked spectra occur within the low-altitude projection of the cusp, polar cap, and low-latitude boundary layer. At midnight, although the ions cool toward the poleward edge of the oval, the ion spectrum is not peaked at low energies as at dawn, dusk, and noon. We suggest that these ions originate in the boundary plasma sheet and that the difference in spectral characteristics reflects different processes operating on the low-energy ions in the low-latitude boundary layer and boundary plasma sheet.

In the average maps the center of the cusp is identified with the integral number flux maximum on the dayside. The cusp so defined is found centered on noon at all levels of activity. The integral flux falls off both in local time and latitude from this maximum. That is, the maximum is cusp-like (narrowly confined in both latitude and MLT) as op-

posed to cleftlike (confined more substantially in latitude than MLT). The electron dayside region of high number flux and low average energy has been shown to be cleftlike [Hardy et al., 1985]. The ion average energy minimum also varies in a cusplike manner with the minimum located along the noon meridian poleward of the number flux maximum. At high latitudes the average energy increases with local time away from noon. The average energy, as a function of MLT and at a fixed latitude within the region of peaked ion spectra, increases to higher values postnoon than prenoon. In terms of the plasma regimes to which this low-altitude regime connects, this implies that the average energy increases and the integral number flux decreases as a function of MLT time away from noon going from the cusp into the closed field lines of the low-latitude boundary layer and the open field lines of the polar cap and both the integral number flux and average energy decrease with latitude across the cusp toward the polar cap. This same behavior was seen in the average patterns for the electrons [Hardy et al., 1985].

The decrease in the average energy across the cusp poleward of the number flux maximum is in agreement with previous reports of cusp energy dispersion produced by the motion of the ions in the convection electric field [Reiff et al., 1977, 1980; Shelley et al., 1979; Burch et al., 1980]. For the entry of ions into the cusp from the magnetosheath the ions with the highest energy will be observed at the lowest latitude, while the lower-energy ions, because of their smaller parallel velocities, will have a longer transit time and will be convected farther poleward. The distance the ions are convected is roughly, directly proportional to the strength of the convection electric field. Since the strength of the electric field increases with increasing activity, the distance between the integral number flux and the average energy minimum should also increase with Kp , as we observe. The difference in latitude increases from 1.5° at $Kp = 0$ to 7° for $Kp \geq 6$, indicating a greater than fourfold increase in the average strength of the convection electric field. Such a range in values is consistent with the measurements of the cross polar cap potential reported in the literature [Burke and Doyle, 1986, and references therein].

Reiff, Burch, and coworkers have also reported satellite passes where the ion average energy increased with latitude across the cusp. They identified such behavior as the signature for diffusion as the primary process for populating the

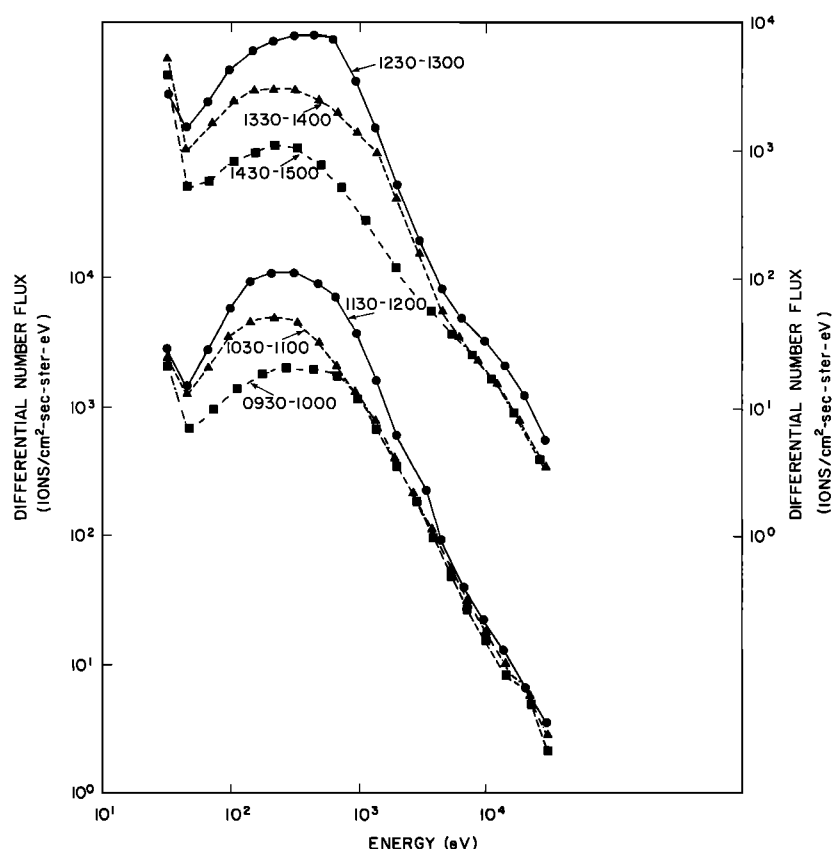


Fig. 18. The logarithm of the differential number flux plotted versus energy on a logarithmic scale for six magnetic local times at the latitude of the integral number flux maximum in the cusp for $Kp = 3$.

cusp at these times. These signatures were observed primarily for a northward B_z component in the IMF. At no activity level do we see any indication in the average patterns of such a signature. Even for the $Kp = 0$ case which should correspond primarily to times when the IMF B_z component is positive, the same decrease in average energy across the cusp is observed. This argues that at all levels of activity such signatures occur rarely and that diffusion on average plays a minor role in populating the cusp.

Sauvaud et al. [1980] have argued that the cusp exists only sporadically on the dayside. They defined the cusp as regions where the ratio of the differential number flux of ions at 500 eV and 5 keV was greater than 100. The apparent sporadic character of the cusp that they report may be the result of this restrictive definition of the cusp. In Figure 18 the ion differential number flux spectra are plotted at 1-hour intervals about noon for the latitude of the center of the cusp for $Kp = 3$. All the spectra are similar in shape with a peak at a few hundred eV. For local time away from noon the intensity at the peak decreases while the intensity of the high energy tail remains approximately the same or, as on the post-noon side, increases slightly. All these spectra occur in the cusp or boundary regions. Only the spectrum at noon, however, meets the criterion as defined by *Sauvaud et al.* [1980] for the cusp. Similarly, the criterion is not satisfied for the softer spectra poleward of the integral flux maximum shown in Figure 7. If the average patterns derived represent the basic pattern encountered on a given pass, then the *Sauvaud et al.* criterion will be satisfied only in a small area and encountering this region will be very much orbit dependent making the cusp appear to exist only sporadically.

We showed in Figure 18 that the increase in the average energy with MLT away from noon results from a decrease in the intensity of the flux at the peak at a few hundred eV, while the tail of the spectrum above 1 keV remained constant or increased. The decrease in the intensity of the peak is expected between open field lines in the cusp and closed field lines of the low-latitude boundary layer. For the case of hydromagnetic flow the density of the magnetosheath will decrease with increasing distance from the nose towards the flanks of the magnetosphere [*Spreiter and Alksne*, 1969]. *Eastman et al.* [1976] and *Eastman and Hones* [1979] have reported that there is no marked change in the plasma characteristics between the magnetosheath and the low-latitude boundary layer. Thus the density in the low-latitude boundary layer should follow the decrease in the density of the magnetosheath resulting in the diminished peak that we see at low altitudes.

In a case of pure hydromagnetic flow, if the behavior of the low-latitude boundary layer follows the behavior of the magnetosheath, then the bulk flow should increase and the temperature decrease in the low-altitude region with increasing MLT away from noon [*Spreiter and Alksne*, 1969]. This is not observed. This can be explained by the fact that if the low-latitude boundary layer lies, partially or wholly, on closed field lines, then it can also be populated by plasma originating in the nightside magnetosphere and transported by the convection electric field and gradient and curvature drifts to the dayside. This plasma can consist either of particles originating in the plasma sheet in the geomagnetic tail or cold particles from the ionosphere injected into the inner magnetosphere. The precipitating ion spectrum, there-

fore, will have contributions from the nightside source and the magnetosheath and the changes due to hydromagnetic flow will be obscured.

The greater increase in the average ion energy at fixed cusp latitude with MLT postnoon than prenoon supports the contention that the spectrum results from multiple sources. Higher-energy ions from the plasma sheet follow trajectories preferentially on the evening flank of the inner magnetosphere. These trajectories tend to intersect the magnetopause in the late morning and afternoon such that they are excluded from much of the morningside oval. As a function of increasing energy, the ion trajectories intersect the magnetopause at progressively later local times [Ejiri, 1978]. Such ions as they are transported to the magnetopause will intersect the low-latitude boundary region. Because the higher-energy ions intersect the magnetopause preferentially postnoon, they will make a larger contribution to the spectra mapping to the eveningside of the low-latitude boundary layer, producing a more prominent high-energy tail, as is observed.

This combination can also explain the asymmetry in the level of the integral number flux about noon. Low-energy ions from the nightside magnetosphere have trajectories on the morningside that connect to the magnetopause prenoon. On the eveningside the low-energy ions have limited access to the postnoon magnetopause since they must pass through the stagnation point [see Ejiri, 1978]. As such, these low-energy ions could make a larger contribution to the morningside than the eveningside region, connecting to the low-latitude boundary layer, possibly raising the level of integral number flux and producing the observed asymmetry.

For open field lines in the plasma mantle or polar cap a similar behavior in the intensity of the peak in the spectrum and the average energy is expected. Open field lines away from noon map to the flanks of the magnetosphere and into the geomagnetic tail. Unlike field lines in the cusp, for these field lines there is no component of the plasma flow along the field line direction into the magnetosphere. As such the particle access is reduced and the peak intensity should decrease. Similarly, high-energy ions can more easily flow back along the field lines in this situation than can the lower-energy ions. For such a situation there is, however, no reason for an asymmetry in the average energy or intensity postnoon and prenoon.

6. CONCLUSION

In conclusion, we have shown that the auroral precipitating ions occur in a well-ordered pattern in magnetic local time and geomagnetic latitude when separated by K_p . A portion of the pattern is found to be qualitatively consistent with the adiabatic compression of the ions as they are convected earthward in the tail and then gradient and curvature drift. The apparent adiabatic behavior of the plasma is clear in the dusk sector and at high latitudes in the dawn sector. At midnight the effects are not as clear due to substorm processes that may violate the adiabatic invariants and that produce uncertainties in mapping auroral measurements into the tail.

The maximum value of the integral energy flux is seen premidnight at all levels of activity. The higher levels of integral energy flux occur in C-shaped regions symmetric about a premidnight-prenoon meridian. The patterns of electron and ion integral energy flux are approximately

symmetric, mirrored about the noon-midnight meridian. The peak energy flux in latitude carried by the hot ions from the plasma sheet decreases by an order of magnitude or more from the premidnight to prenoon sectors for high activity.

The mapping to low altitudes of the cusp, plasma mantle and low-latitude boundary is identified with a region containing ion spectra peaked at a few hundred eV. The center of the cusp is identified with the maximum integral number flux of such spectra and is centered at noon for all levels of activity. The highest levels of integral number flux on the dayside are confined in latitude and MLT to a relatively small, oval-shaped region consistent with the mapping to low altitudes of a funnel-shaped magnetic field region. The position of the center of the cusp moves equatorward with increasing geomagnetic activity by 10° from $K_p = 0$ to $K_p \geq 6-$. Poleward of the integral number flux maximum the average energy decreases to a minimum consistent with the convection electric field operating as a velocity filter for cusp ions that originate in the magnetosheath. The latitude of the average energy minimum shows a smaller equatorward motion with increasing activity than the latitude of the number flux maximum. With local time away from noon the integral number flux decreases, and the average energy of the ions increases for field lines that thread the low-latitude boundary layer and the plasma mantle. This behavior is consistent with these regions being populated by a combination of plasma originating in the magnetosheath and penetrating across the magnetopause on the dayside and plasma originating in the tail and nightside magnetosphere and transported to the dayside.

The integral number flux of the ions is found to be 1–2 orders of magnitude smaller than that for electrons at all levels of activity and all latitudes and local times. The integral energy flux for the ions can equal or exceed that for the electrons for some latitudes and local times on the eveningside of the oval and for much of the oval is a significant fraction of the electron integral energy flux. Integrated over a hemisphere, the number flux of the ions varies from 3.62×10^{23} to 1.5×10^{24} ions/s sr from $K_p = 0$ to $K_p \geq 6-$. As a percentage of the hemispheric electron number flux, the ions decrease from 2.37% at $K_p = 0$ to 1.47% for $K_p \geq 6-$. The hemispheric energy flux of ions varies from 7.26×10^8 W at $K_p = 0$ to 1.00×10^{10} W for $K_p \geq 6-$. The hemispheric energy input from the ions varies between 11 and 17% of the energy input for the electrons and shows no trend with activity.

APPENDIX

The following describes the technique and justification for the extrapolation of the average ion spectra. In Figure A1 the logarithm of the distribution function is plotted versus energy on a linear scale for the average ion spectra at a set of latitudes for the 0000–0030 and 1800–1830 MLT zones and for $K_p = 3$. The spectra are broadly representative of spectra at other latitudes, MLTs and K_p s. In Figure A1 the eight highest-energy channels are plotted. In such a format a Maxwellian distribution appears as a straight line with a slope inversely proportional to the temperature and an intercept proportional to the number density. In Figure A1 the slope of the curves decreases with increasing energy. In general, however, the four highest-energy channels can be well approximated by a straight line. Using this fact, we performed a linear least squares fit to the four highest-energy

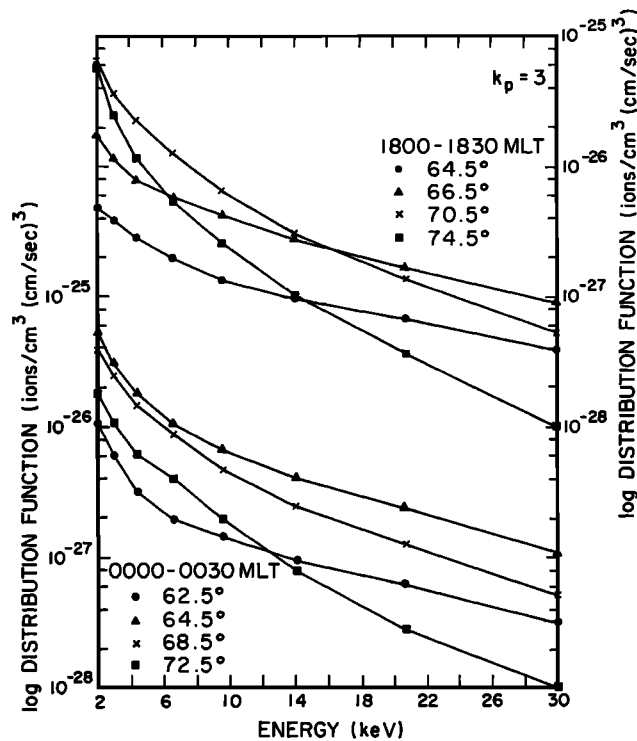


Fig. A1. The logarithm of the average ion distribution function plotted versus energy on a linear scale for $K_p = 3$ and for four latitudes at midnight and 1800 MLT. The scale for the midnight spectra is on the left-hand side and for the 1800 MLT spectra the scale is on the right-hand side.

channels and then used this fit to extrapolate the flux level and the count rate for three additional channels at approximately 44, 64, and 94 keV. The channels were chosen to have the same logarithmic spacing in energy as the real channels of the detector. Since the slope of the curves in Figure A1 tends to decrease with increasing energy, the technique most likely underestimates the contribution from the high-energy portion of the spectrum. Spectra were extrapolated only if the average counts in the 30-keV channel were greater than 1 count per accumulation interval. This level was chosen to avoid extrapolating any small amount of noise remaining in the average spectra.

In Figure A2, three examples of spectra extrapolated in this manner are plotted. In Figure A2 the extrapolated portion of each spectrum is denoted with a dashed curve. One notes that the extrapolation provides a reasonable continuation of the spectral shape measured at lower energies. The unextrapolated and extrapolated values of the integral energy flux, integral number flux, and average of the three spectra are listed in Table A1. The three spectra are representative of the minority of average spectra where the extrapolation has a major effect. For the three cases the integral energy flux is increased by 53–187%, the integral number flux by 13–52%, and the average energy by 32–89%. The majority of this increase comes from the first extrapolated channel. For the spectrum at 0000–0030, 1800–1830, and 1900–1930 respectively the addition of the first channel accounts for 63, 69, and 41% of the increase of the integral energy flux, the second channels 31, 27, and 38% and the last channel 6, 4, and 21%. For the integral number flux and average energy the relative contribution of the two higher-energy channels is much smaller.

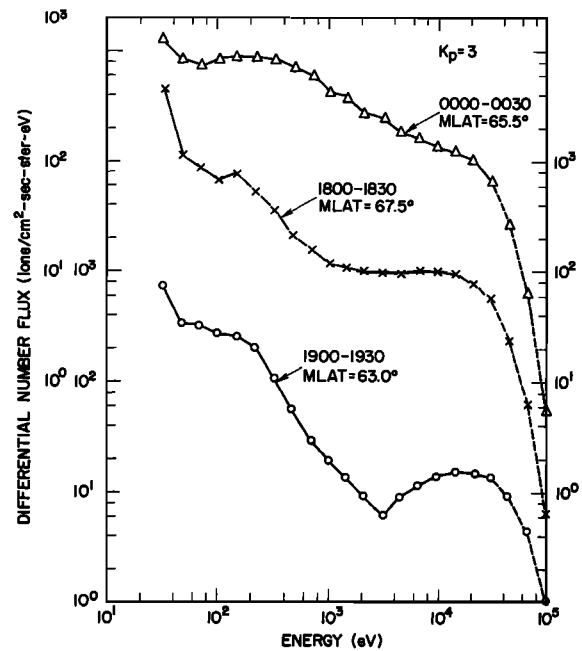


Fig. A2. Representative ion differential number flux spectra plotted versus energy for $K_p = 3$. These average spectra have been extrapolated to 100 keV.

The three examples illustrate how the extrapolation results in values of the average energy more consistent with the measured portion of the spectrum than the values calculated from the unextrapolated spectra. In Figure A2 the differential number flux spectra at 1800 and 1900 MLT are approximately Maxwellian in shape for energies above a few keV. At lower energies the flux increases monotonically for decreasing energy. Since the differential number flux spectrum of a Maxwellian distribution peaks at an energy equal to the temperature, these two spectra should have calculated temperatures of approximately 10 and 15 keV. Since our calculated average energy is twice the temperature, the two distributions should have average energies of approximately 20 and 30 keV. Referring to Table A1, one notes that the average energies from the extrapolated spectra are much closer to these values than the average energies from the unextrapolated spectra.

Three additional points relative to the extrapolation are as follows: (1) Over most of the high-latitude region the effect of the extrapolation is negligible. The effect is important primarily in the midnight and evening sectors towards the equatorward edge of the oval and is small in the dayside and morning sectors and in the polar cap and boundary regions. (2) For low activity, $K_p = 0$ and 1, the effect is small at all latitudes and local times. (3) The overall effect increases with increasing activity.

TABLE A1. Unextrapolated and Extrapolated Integral Quantities

	EAVE	JTOT	JETOT
1900–1930	15.9	5.97×10^5	9.50×10^6 (unextrapolated)
(63.0°)	30.2	9.05×10^5	2.73×10^7 (extrapolated)
1800–1830	14.8	3.20×10^6	4.74×10^7 (unextrapolated)
(67.5°)	19.6	4.04×10^6	7.91×10^7 (extrapolated)
0000–0030	12.0	5.21×10^6	6.24×10^7 (unextrapolated)
(65.5°)	16.3	5.86×10^6	9.54×10^7 (extrapolated)

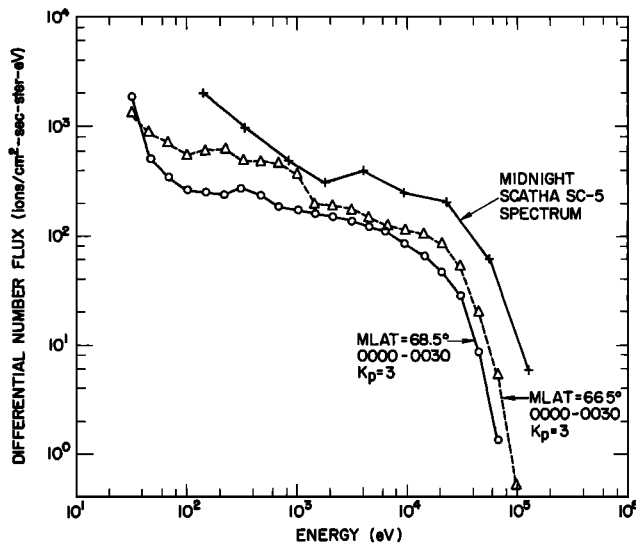


Fig. A3. The average ion spectrum at the SCATHA satellite orbit in the midnight sector, perpendicular to the magnetic field and averaged over all activity compared to two average spectra in the auroral zone in the 0000–0030 MLT zone for $K_p = 3$.

As a final check on the validity of the extrapolation, average ion spectra in the midnight sector from the DMSP and SCATHA satellites were compared (Figure A3). The SCATHA satellite was flown in a near-equatorial orbit spanning the L shells from approximately 5.5 to 7.5. The SCATHA spectrum was calculated from the SC5 experiment by averaging the ion spectra taken perpendicular to the magnetic field in the midnight sector, over all activity and over all L shell sampled [Mullen and Gussenhoven, 1983]. The SC5 detector measured ions in the energy range from approximately 150 eV to greater than 100 keV. For comparison the DMSP spectra are taken in the 0000–0030 MLT zone for $K_p = 3$ and at latitudes of 66.5° and 68.5°. These latitudes should bracket the L shell sampled by SCATHA. The $K_p = 3$ case was chosen since the long-term average of K_p has a value between 2 and 3. The spectra are seen to agree well over the entire energy range, again suggesting that the extrapolation produces reasonable results. The level of flux differs by a factor of 2–3 which we attribute primarily to the spectrum being measured perpendicular to the field line at SCATHA and essentially parallel to the field line at DMSP.

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